

NOZZLE SPARGER

Most modern mechanically stirred fermenter designs from laboratory to industrial scale have a single open or partially closed pipe as a sparger to provide the stream of air bubbles. Ideally the pipe should be positioned centrally below the impeller and as far away as possible from it to ensure that the impeller is not flooded by the air stream (Finn, 1954). The single-nozzle sparger causes a lower pressure loss than any other sparger and normally does not get blocked.

COMBINED SPARGER-AGITATOR

On a small scale (1 dm³), Herbert *et al.* (1965) developed the combined sparger-agitator design, introducing the air via a hollow agitator shaft and emitting it through holes drilled in the disc between the blades and connected to the base of the main shaft. The design gives good aeration in a baffled vessel when the agitator is operated at a range of rpm (see Fig. 7.17).

THE ACHIEVEMENT AND MAINTENANCE OF ASEPTIC CONDITIONS

Once the design problems of aeration and agitation have been solved, it is essential that the design meets the requirements of the degree of asepsis and containment demanded by the particular process being considered. It will be necessary to be able to sterilize, and keep sterile, a fermenter and its contents throughout a complete growth cycle. There may be also a need to protect workers and the environment from exposure to hazardous micro-organisms or animal cells (Van Houten, 1992). As has been mentioned earlier, the containment requirements depend on the size of the fermentation vessel.

The following operations may have to be performed according to certain specifications to achieve and maintain aseptic conditions and containment during a fermentation:

1. Sterilization of the fermenter.
2. Sterilization of the air supply and the exhaust gas.
3. Aeration and agitation.
4. The addition of inoculum, nutrients and other supplements.
5. Sampling.
6. Foam control.
7. Monitoring and control of various parameters.

On a small scale, below 10 dm³, the biohazard risk can be controlled by a combination of containment cabinets and work practices (Van Houten, 1992). When the volume of culture exceeds 10 dm³, GILSP is required for those non-pathogenic non-toxic agents which have an extended history of large scale use. For this category there should be prevention of contamination of the product, control of aerosols and minimization of the release of micro-organisms during sampling, addition of material, transfer of cells and removal of materials, products and effluents. It should be appreciated that the majority of fermentations fall into this category.

At level 1, B2 or at higher containment levels, the following points need to be considered when designing a fermenter or other vessel, so that it can operate as a contained system (Tubito, 1991):

1. All vessels containing live organisms should be suitable for steam sterilization and have sterile vent filters. This is discussed in Chapter 5.
2. Exhaust gases from vessels should pass through sterile filters. This is discussed in Chapter 5.
3. Seals on flange joints should be fitted with a single 'O'-ring at the lower levels of containment. Flange joints on vessels for Containment levels 3 and B3/4 need double 'O'-rings or double 'O'-rings plus a steam barrier. This has been discussed in an earlier section of this chapter.
4. Appropriate seals should be provided for entry ports for sensor probes, inoculum, sampling, medium addition, acid, alkali and antifoam. This will be discussed later in this chapter.
5. Rotating shafts into a closed system should be sealed with a double acting mechanical seal with steam or condensate between the seals. This has been discussed in an earlier section in this chapter.
6. During operation a steam barrier should be maintained in all fixed piping leading to the 'contained' vessels.
7. Provision of appropriate pressure relief facilities will be discussed later in this chapter.

Further details for containment are given by Giorgio and Wu (1986), Hesselink *et al.* (1990), Kennedy *et al.* (1990), Janssen *et al.* (1990), Hambleton *et al.* (1991), Tubito (1991), Leaver and Hambleton (1992), Van Houten (1992), Vranich (1992) and Werner (1992).

Some of the issues discussed in points 1, 3, 4, 5 and 7

will need to be considered for aseptic operation of GILSP processes.

Sterilization of the fermenter

The fermenter should be so designed that it may be steam sterilized under pressure. The medium may be sterilized in the vessel or separately, and subsequently added aseptically. If the medium is sterilized *in situ* its temperature should be raised prior to the injection of live steam to prevent the formation of large amounts of condensate. This may be achieved by steam being introduced into the fermenter coils or jacket. As every point of entry to and exit from the fermenter is a potential source of contamination, steam should be introduced through all the entry and exit points except the air outlet from which steam should be allowed to leave.

All pipes should be constructed as simply as possible and slope towards drainage points to make sure that steam reaches all parts of the equipment and is not excluded by siphons or pockets of condensate or mash. Each drainage point in the pipework should be fitted with a steam trap. This will be described in the section on valves and steam traps. Parker (1958), Chain *et al.* (1954) and Muller and Kieslich (1966) and others have all stressed the need to eliminate fine fissures or gaps such as flange seals which might be filled with nutrient solutions and micro-organisms. Hambleton *et al.* (1991) described a high specification pilot scale fermenter with surfaces free of crevices greater than 0.05-mm depth, which is needed if the vessel was to be used for animal cells in suspension culture or on micro-carriers. For long-term aseptic operation welded joints should be used wherever possible, even though sections may have to be cut out and re-welded during maintenance and repair (Smith, 1980).

Sterilization of the air supply

Sterile air will be required in very large volumes in many aerobic fermentation processes. Although there are a number of ways of sterilizing air, only two have found permanent application. These are heat and filtration. Heat is generally too costly for full-scale operation (see also Chapter 5).

Historically, glass wool, glass fibre or mineral slag wool have been used as filter material, but currently most fermenters are fitted with cartridge-type filters as discussed in Chapter 5. However, before the filter may

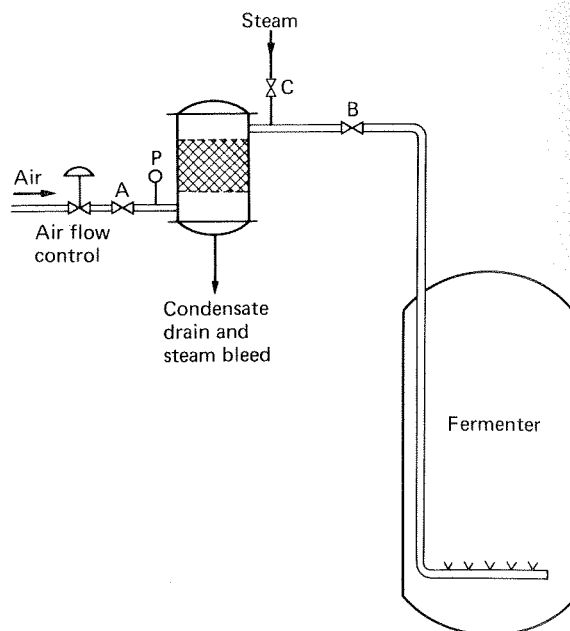


FIG. 7.18. An arrangement of packed air filter and fermenter (Richards, 1968).

be used it, too, must be sterilized in association with the fermenter. Two procedures are commonly followed depending on the construction of the filter unit.

Figure 7.18 shows the simple unit described by Richards (1968). During sterilization the main non-sterile air-inlet valve A is shut, and initially the sterile air valve B is closed. Steam is applied at valve C and air is purged downwards through the filter to a bleed valve at the base. When the steam is issuing freely through the bleed valve, the valve B is opened to allow steam to pass into the fermenter as well as the filter. It is essential to adjust the bleed valve to ensure that the correct sterilization pressure is maintained in the fermenter and filter for the remainder of the sterilization cycle.

An alternative approach is to use a steam-jacketed air filter (Fig. 7.19). At the beginning of a sterilization cycle the valve A will be closed and steam passed through valves B and C, and bled out of D. Simultaneously steam will be passed into the steam jacket through valve F and out of G. When steam is issuing freely from valve D, valve F, may be opened and steam circulated into the fermenter. The bleed valve D will have to be adjusted to ensure that the correct pressure is maintained. Once the sterilization cycle is complete, valves B and E are closed and A is opened to allow air to pass through the heated filter and out of valve D to

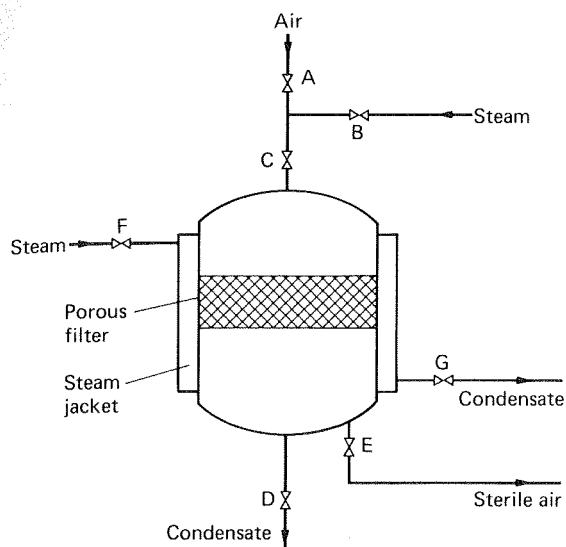


FIG. 7.19. Design for a simple steam-jacketed air-filter.

dry the filter. Finally the steam supply to the steam jacket is stopped. Valve D is closed and valve E opened, thus introducing sterile air into the fermenter to achieve a slight positive pressure in the vessel.

Sterilization of the exhaust gas from a fermenter

Sterilization of the exhaust gas can be achieved by 0.2- μm filters on the outlet pipe (Fig. 5.21). Under normal operation aerosol formation may occur in the fermenter and moisture and solid matter may then plug the filter. To ensure satisfactory operation a cyclone separator (for solids) and a coalescer (for liquids) would be included upstream of two filters in series. The filters should be checked regularly to ensure that no viable cells are escaping. A test procedure to ensure integrity has been described by Hesselink *et al.* (1990).

The addition of inoculum, nutrients and other supplements

To prevent contamination when operating a fermenter requiring GILSP, it is essential that both the addition vessel and the fermenter should be maintained at a positive pressure and that the addition port is equipped with a steam supply.

At Containment levels 1 and B2, the addition of inoculum, nutrients, etc. must be carried out in such a way that release of micro-organisms is restricted. This

should be done by aseptic piercing of membranes or connections with steam locks. At Containment levels 2 and B3/4, no micro-organisms must be released during inoculation or other additions. In order to meet these stringent requirements all connections must be screwed or clamped and all pipelines must be steam sterilizable (Werner, 1992).

Further details of the aseptic inoculation of laboratory, pilot-plant and production fermenters are described in Chapter 6.

Sampling

The sampling points fitted to larger fermenters also illustrate the principles for maintaining sterility. A sterile barrier must be maintained between the fermenter contents and the exterior when the sample port is not being used and it must be sterilizable after use. A simple design (Fig. 7.20) is described by Parker (1958). In normal operation valves A, B and C are closed and a barrier is formed by submerging the end of the sampling port in 40% formalin or a suitable substitute. A sample is obtained by removing the container of formalin and closing valve A. Valves B and C are then opened until the piping has been sterilized by steam. Valves B and C are then partially closed to allow a slow stream of steam and condensate out of the sampling port. Valve A is then opened slightly to cool the piping. The broth is discarded. Valve C is then closed and a sample is collected. Valve A is then closed and the piping is resterilized and left in the out of use arrangement.

An alternative arrangement for a sampling port is illustrated in Fig. 7.21 where the sterile barrier between

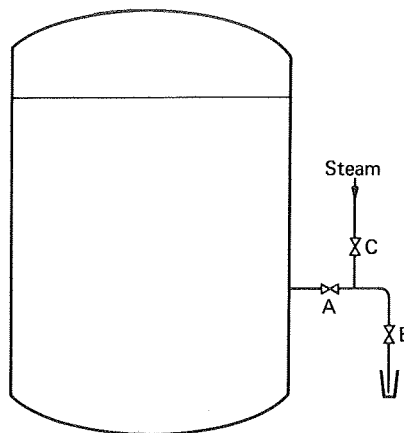


FIG. 7.20. Simple design for a sampling port (Parker, 1958).

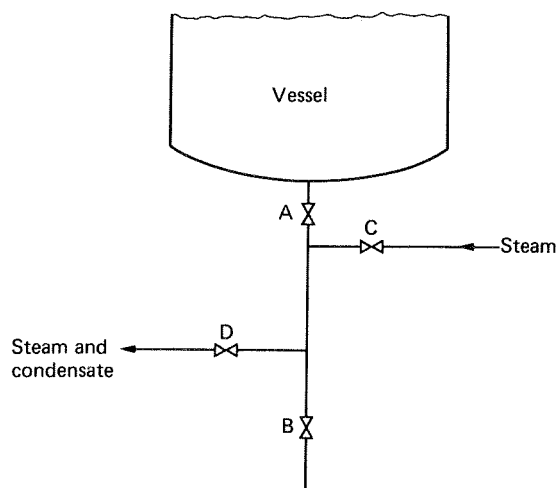


FIG. 7.21. An alternative simple sample port.

the sample port and the exterior is a condensed flow of steam between valves C and D, valves A and B being closed. Valve D is connected to a steam trap to avoid condensate accumulation. To sterilize valve B prior to sampling, valve C is partially closed, valve D completely closed and valve B partially opened to allow a slow stream of steam and condensate out of the sampling port. Valve A is opened briefly to cool the pipe and the broth is discarded. Valve C is then closed and a sample is collected. Valve A is then closed and the piping is resterilized. In between collecting samples valves C and D are left partially open.

A more modern sterilizable system, as illustrated in Fig. 7.22, has been described by Werner (1992). In the closed position steam enters through the entry point 6 and passes through the sampling hole 11 and into the sampling hole 2. When sampling, the handle is pushed so that the hole in the sampling tube is in the fermenter and the vessel contents may be sampled. When the sampler is closed the unit can be resterilized.

Marshall *et al.* (1990) recognized that sampling requires good mechanical design and good operator practice to ensure sterility. Complex valve sequencing can lead to operator error. The automatic MX-3 rotary valve (Fig. 7.23) was developed by Marshall Biotechnology Ltd and is now marketed by New Brunswick Scientific Ltd. In this device, broth is continuously recycled to and from the fermenter and up to 12 samples can be taken automatically in sample bottles according to a pre-programmed sequence. The storage of the bottles in an integral refrigeration block re-

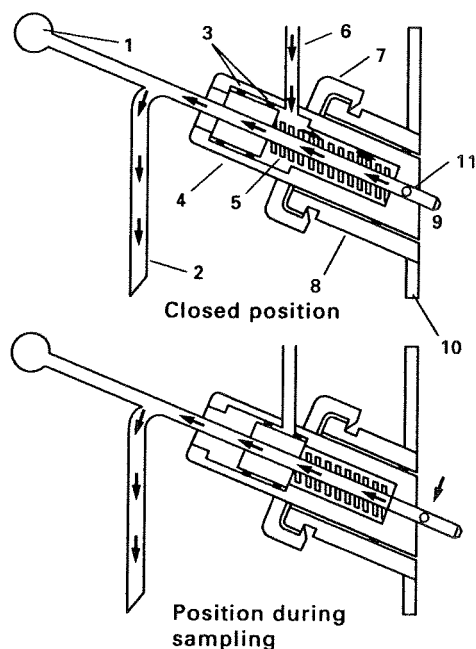


FIG. 7.22. Sterilizable sampling system for Category 1 (Werner, 1992). Key: (1) handle for opening and closing; (2) piping for sample or condensate, respectively; (3) 'O'-ring; (4) housing; (5) spring; (6) steam inlet; (7) union nut; (8) welding socket; (9) product; (10) wall of the bioreactor; (11) port of sampling tube.

duces spoilage until they can be removed for analysis. This sampler can be used for GILSP.

In all the sampling devices described above, steam is used merely to maintain aseptic conditions. If the process organism has to be contained then the sample vessel must be designed accordingly. At level 1 or B2, removal of broth should be carried out in such a way that release of micro-organisms is minimized. Such a system (Fig. 7.24) is described by Janssen *et al.* (1990). The sampling bottle assembly is attached to the fermenter by a double-end shut-off quickconnect B and air is vented from the vessel during sampling by a suitable membrane filter. The pipe work is steamed for 15 minutes using 1.3 bar steam with the valve E on the sample bottle closed. With valves C and D closed and A and B open a sample can be taken, then the pipe is resterilized before removing the sample bottle.

At level 2 or B3, sampling should be done in a closed system (Figs 7.25 and 7.26), and all piping coming into contact with a sample must be sterilizable with steam. The sampling vessel should be closed during transport and samples should be examined under containment

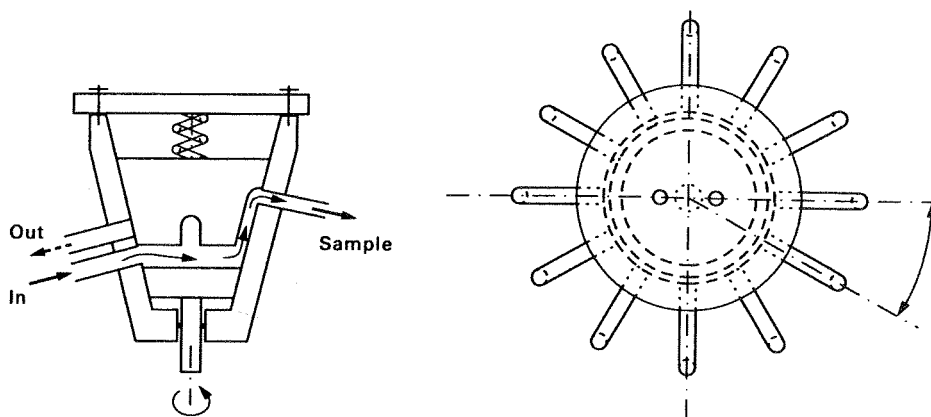


FIG. 7.23. Schematic representation of the MX-3 BioSampler rotary valve system.

conditions corresponding to those specified for the process (Werner, 1992).

intermittently (Fig. 7.27). These can be sterilized *in situ* with steam after connection has been completed and before any additions are made.

Feed ports

Additions of nutrients and acid/alkali to small fermenters are normally made via silicone tubes which are autoclaved separately and pumped by a peristaltic pump after aseptic connection. In large fermenter units, the nutrient reservoirs and associated piping are usually an integrated part which can be sterilized with the vessel. However, there may also be ports which are used

Sensor probes

Double 'O' ring seals have been used for many years to provide an aseptic seal for glass electrodes in stainless steel housings in fermenters using GILSP. This system is also suitable for levels 1 and B2, provided that release of micro-organisms is minimized and there are adequate disinfection procedures for dealing with leakages (Werner, 1992).

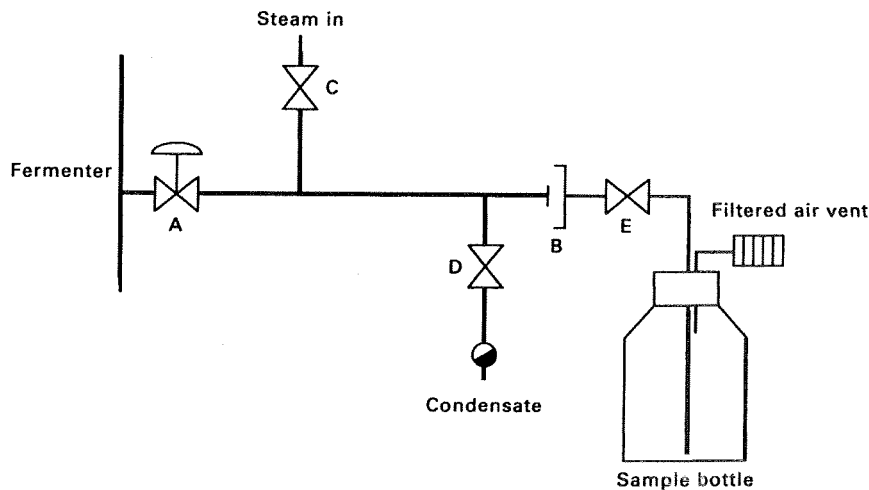
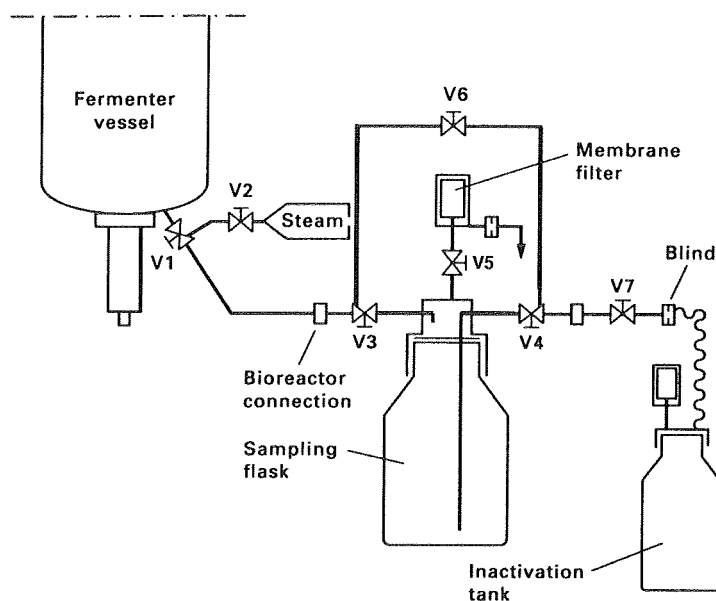


FIG. 7.24. Sample system for level 1 (Jannsen *et al.*, 1990).



Valve sequences during sampling

Valve	Sterilization sampling flask	Sampling	Sterilization after sampling of sampling flask	Removal of sampling flask
V1	closed	open	closed	closed
V2	open	closed	open	closed
V3	open	open	closed	closed
V4	open	closed	closed	closed
V5	open	open	open	closed
V6	closed	closed	open	closed
V7	open	open	open	closed

FIG. 7.25. Containment sampling unit for Chapter 2 (Werner, 1992).

At Containment levels 2 and B3/4, probes are fitted with triple 'O'-ring seals (Hambleton *et al.*, 1991). Although double 'O'-ring seals with steam tracing have been described they are not normally considered to be feasible (Leaver and Hambleton, 1992). The use of pre-inserted back-up probes is recommended as a means for dealing with probe failure rather than using a retractable electrode housing during a fermentation cycle because of the danger of leakage of broth (Hambleton *et al.*, 1991).

Foam control

In any fermentation it is important to minimize foaming. When foaming becomes excessive, there is a danger that filters become wet resulting in contamina-

tion. There is also the possibility that siphoning will develop, leading to the loss of all or part of the contents of the fermenter. Methods for foam control are considered in Chapter 8 and antifoams are discussed in Chapter 4. In certain circumstances antifoams may cause problems with aeration or downstream processing. Foam breakers which break down foam by an impact mechanism created by some type of rotating mechanism inside the fermenter are manufactured by Bioengineering AG (Switzerland), Chemap AG (Switzerland), Electrolux (Sweden), Frings (Germany) and New Brunswick Scientific Ltd (U.S.A.). Mechanical foam breakers used in Acetators for vinegar production are described later in this chapter. Domnick Hunter (U.K.) manufacture the 'Turbosep', in which foam is directed over stationary turbine blades in a separator and the liquid fraction is returned to the fermenter.

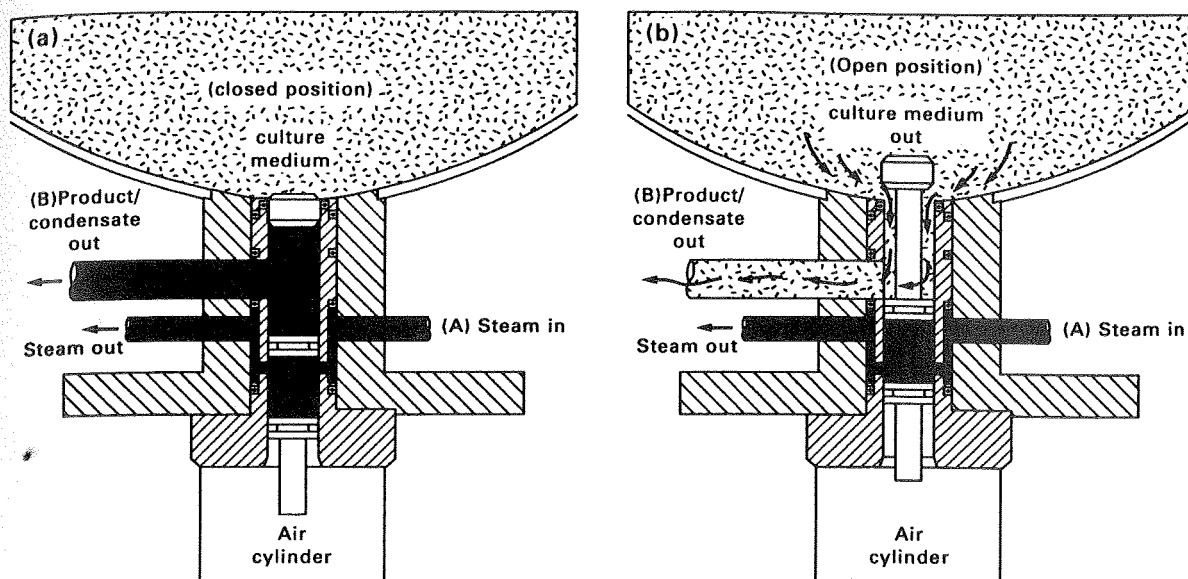


FIG. 7.26. Resterilizable harvest-sampling valve for Level 2 containment (New Brunswick, Hatfield, England). This spool-type valve is connected to the bottom of the fermenter vessel in the closed position (a), pressurized steam circulates throughout the entire valve body and through the product condensate line (B) via a steam inlet line (A). Aseptic withdrawal of samples is achieved with the valve in the open position (b). To prevent possible contamination when the plunger is raised, steam is circulated to the lower valve area. Action of the plunger is controlled by an air cylinder.

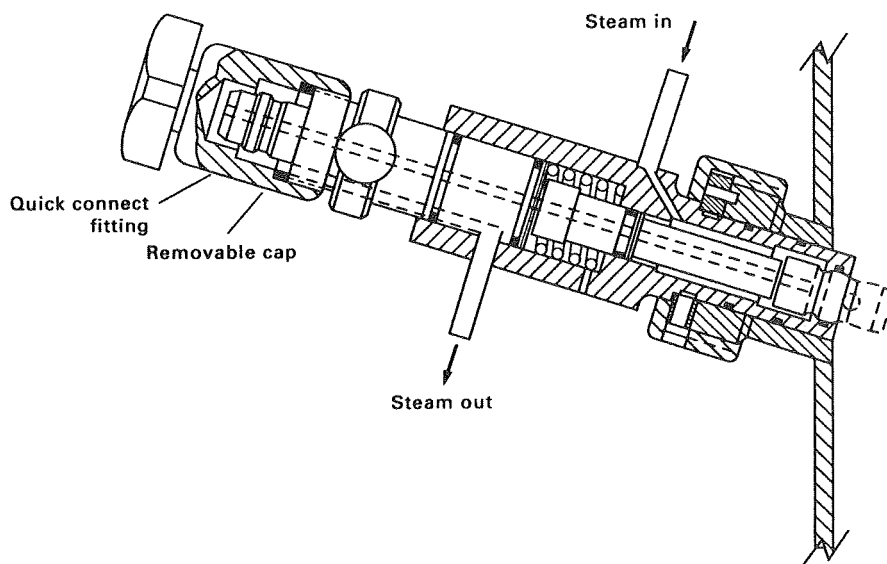


FIG. 7.27. Steam sterilizable liquid-feed port (Beck *et al.*, 1987).

At Containment level 1 or B2, the exhaust air system should shut down automatically if the foam control system fails (Werner, 1992).

Monitoring and control of various parameters

These factors are considered in detail in Chapter 8.

VALVES AND STEAM TRAPS

Valves attached to fermenters and ancillary equipment are used for controlling the flow of liquids and gases in a variety of ways. The valves may be:

1. Simple ON/OFF valves which are either fully open or fully closed.
2. Valves which provide coarse control of flow rates.
3. Valves which may be adjusted very precisely so that flow rates may be accurately controlled.
4. Safety valves which are constructed in such a way that liquids or gases will flow in only one direction.

When making the decision as to which valves to use in the design and construction of a fermenter it is essential to consider the following points:

1. Will the valve serve its chosen purpose? Is it suitable for aseptic operation or contained processes, and of the correct dimensions? Is the pressure drop across the valve tolerable?
2. Will the valve withstand the rigours of the process? The materials used to construct the valve should be suited to the process. It is also important to know whether corrosive liquids are used or synthesized during the process. The maximum operating temperature and pressure of the process should be known.
3. Are there welds or flanges in the valve?
4. Is the valve one which can be operated by remote control?
5. The cost and availability of suitable valves.
6. Can the valve be used for containment purposes?

A wide range of valves is available, but not all of them are suitable for use in fermenter construction (Solomons, 1969; Kemplay, 1980).

The valves described in this section open and close by (a) raising or lowering the blocking unit with a screw

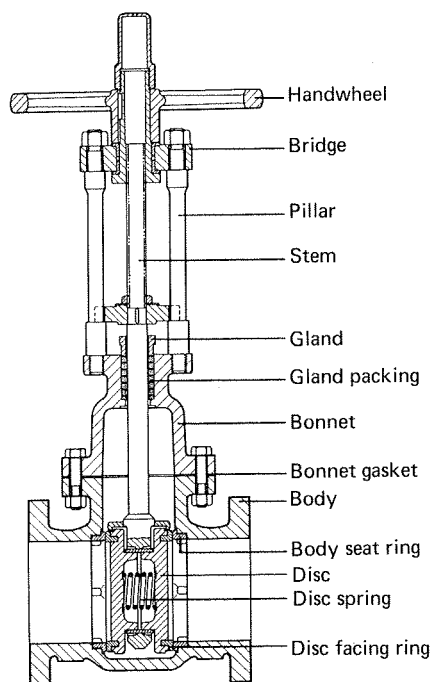


FIG. 7.28. Sectional view of a two-piece gate valve (British Valve Manufacturers Association, 1972).

thread (rising stem), (b) a drilled sphere or plug, or a disc rotating in between two bearings or (c) a rubber diaphragm or tube which is pinched.

Gate valves

In this valve (Fig. 7.28), a sliding disc is moved in or out of the flow path by turning the stem of the valve. It is suitable for general purposes on a steam or a water line for use when fully open or fully closed and therefore should not be used for regulating flow. The flow path is such that the pressure drop is minimal, but unfortunately it is not suitable for aseptic conditions as mash solids can pack in the groove where the gate slides, and there may be leakage round the stem of the valve which is sealed by a simple stuffing box. This means that the nut around the stem and the packing must be checked regularly.

Globe valves

In this valve (Fig. 7.29), a horizontal disc or plug is raised or lowered in its seating to control the rate of

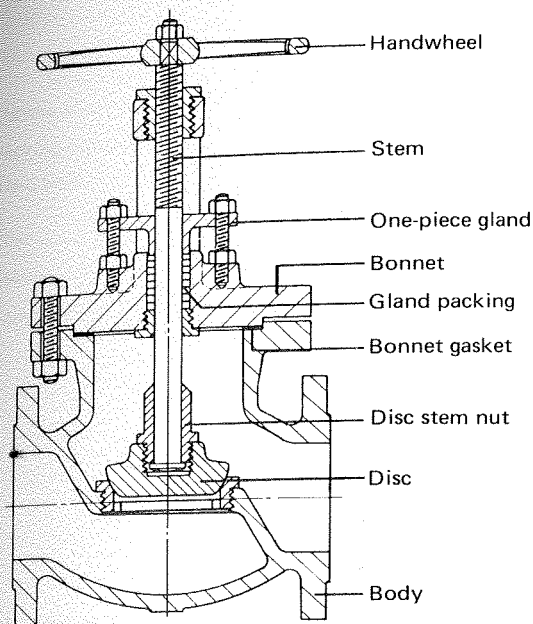


FIG. 7.29. Globe valve with outside screw and conventional disc (Kemplay, 1980).

flow. This type of valve is very commonly used for regulating the flow of water or steam since it may be adjusted rapidly. It is not suitable for aseptic operation because of potential leakage round the valve stem, which is similar in design to that of the gate valve. There is a high-pressure drop across the valve because of the flow path.

In both the gate and globe valves it is possible to incorporate a flexible metallic membrane around the stem of the valve, to replace the standard packing. This modified type of valve can be operated aseptically, but is bigger and more expensive. Valves with non-rising stems have been used, but they are still potential sources of contamination.

Piston valves

The piston valve (Fig. 7.30) is similar to a globe valve except that flow is controlled by a piston passing between two packing rings. This design has proved in practice to be very efficient under aseptic operation. It is important to sterilize them partly open so that steam can reach as far as possible into the valve body. There may be blockage problems with mycelial cultures. The pressure drop is similar to that of a globe valve.

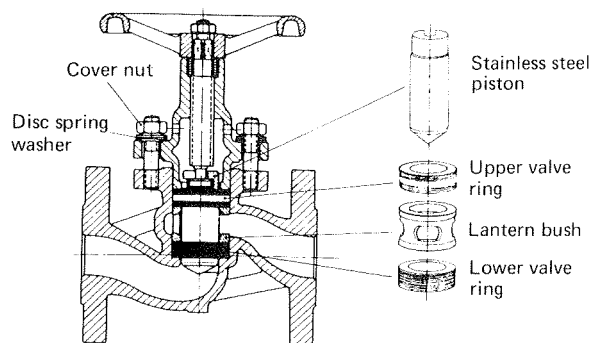


FIG. 7.30. Piston valve (Kemplay, 1980).

Needle valves

The needle valve (Fig. 7.31) is similar to the globe valve, except that the disc is replaced by a tapered plug or needle fitting into a tapered valve seat. The valve can be used to give fine control of steam or liquid flow. Accurate control of flow is possible because of the variable orifice formed between the tapered plug and the tapered seat. The aseptic applications are very limited.

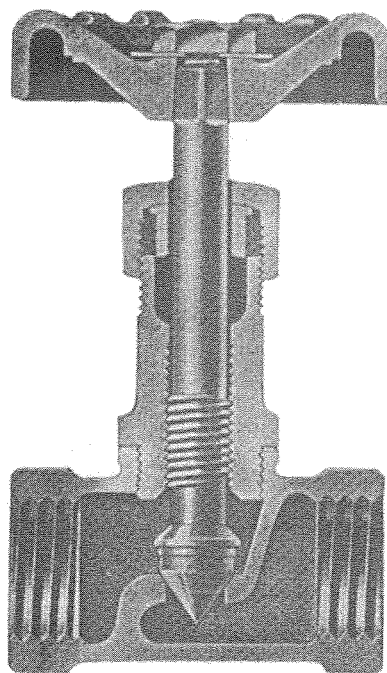


FIG. 7.31. Needle valve for accurate control of flow rate (Kemplay, 1980).

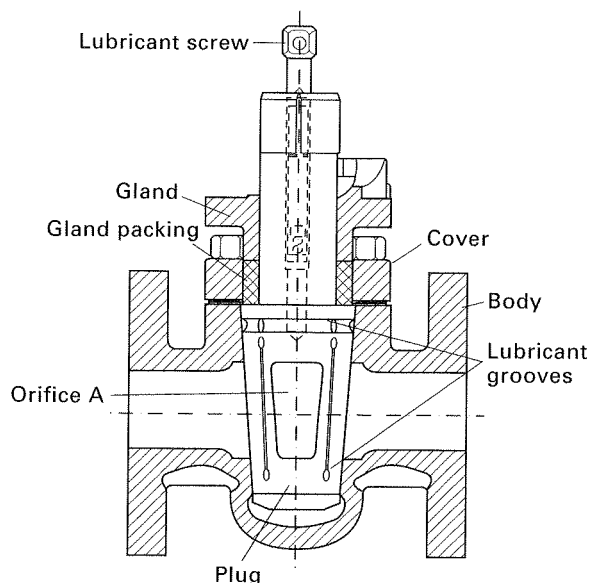


FIG. 7.32. Sectional view of lubricated taper plug valve (British Valve Manufacturers Association, 1972).

Plug valves

In this valve there is a parallel or tapered plug sitting in a housing through which an orifice, A, has been machined. The valve shown in Fig. 7.32 is in the closed position. When the plug is turned through 90° the valve is fully open and the flow path is determined by the cross-sectional area of A, which may not be as large as that of the pipeline. This type of valve has a tendency to leak or seize up, but the use of lubricants and/or sealants may overcome these problems. If suitable packing sleeves (e.g. compressed asbestos) are incorporated into the valve it will be suitable for use in a steam line as it is quick to operate, has protected seals, a minimal pressure drop and a positive closure. It can also provide good flow control.

Ball valves

This valve (Fig. 7.33) has been developed from the plug valve. The valve element is a stainless-steel ball through which an orifice is machined. The ball is sealed between two wiping surfaces which wipe the surface and prevent deposition of matter at this point. The orifice in the ball can be of the same diameter as the pipeline, giving an excellent flow path. The valve is suitable for aseptic operation, can handle mycelial broths and can be operated under high temperatures and pressures. The pressure and temperature range is normally limited by the PTFE seat and stem seals.

Butterfly valves

The butterfly valve (Fig. 7.34) consists of a disc which rotates about a shaft in a housing. The disc closes against a seal to stop the flow of liquid. This type of valve is normally used in large diameter pipes operating under low pressure where absolute closure is not essential. It is not suitable for aseptic operation.

Pinch valves

In the pinch valve (Fig. 7.35) a flexible sleeve is closed by a pair of pinch bars or some other mechanism which can be operated by compressed air remotely or automatically. The flow rate can be controlled from 10 to 95% of rated flow capacity (Pikulik, 1976). The valve is suitable for aseptic operation with fermentation broths, even when mycelial, as there are no dead spaces in the valve structure, and the closing mechanism is isolated from the contents of the piping. Obviously, the sleeve of rubber, neoprene, etc., must be checked regularly for signs of wear.

Diaphragm valves

Like the pinch valve, the diaphragm valve (Fig. 7.36) makes use of a flexible closure, with or without a weir. It may also be fitted with a quick action lever. This

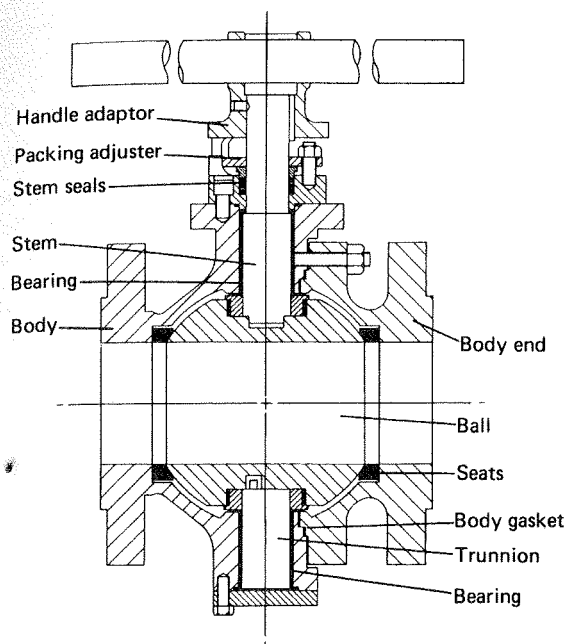


FIG. 7.33. Sectional view of end-entry ball valve (British Valve Manufacturers Association, 1972).

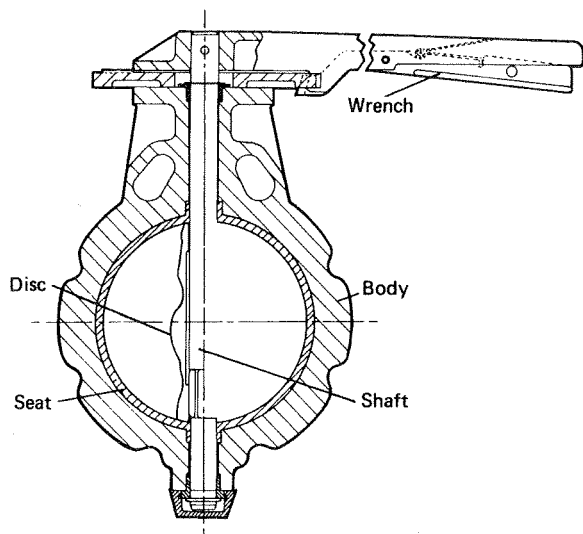


FIG. 7.34. Sectional view of wafer-pattern butterfly valve (British Valve Manufacturers Association, 1972).

valve is very suitable for aseptic operation provided that the diaphragm is of a material which will withstand repeated sterilization. The valve can be used for ON/OFF, flow regulation, and for steam services within

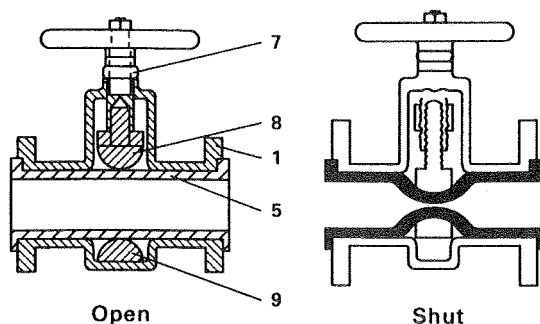


FIG. 7.35. Sectional view of pinch valve in open and shut position: (1) body; (5) flexible tube; (7) spindle; (8) top pinch bar; (9) lower pinch bar (British Valve Manufacturers Association, 1966; Kemplay, 1980).

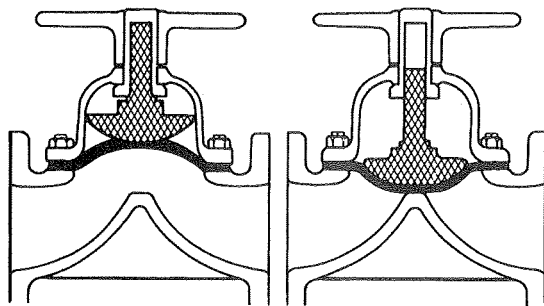


FIG. 7.36. Sectional views of weir-type diaphragm valves in open and closed positions (Thiellsch, 1967).

pressure limits. Diaphragm failure, which is often due to excessive handling, is the primary fault of the valve. Ethylene propylene diene modified (EPDM) is now the preferred material. Hambleton *et al.* (1991) consider that a diaphragm valve with a steam seal on the 'clean' side (APV Ltd, Crawley, U.K.) is a potentially safer valve. However their widescale use would make a fermentation plant much more complex and expensive. Steam barriers on valves have been used by ICI plc for the 'Pruteen' air-lift fermenter (Smith, 1980; Sharp, 1989).

The most suitable valve

Among these group of valves which have just been described, globe and butterfly valves are most commonly used for ON/OFF applications, gate valves for crude flow control, needle valves for accurate flow control and ball, pinch or diaphragm valves for all sterile uses.

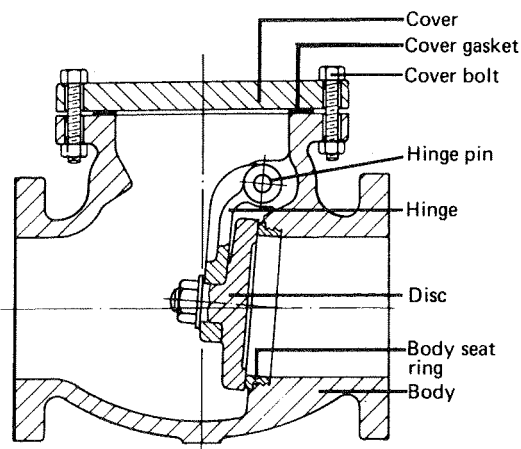


FIG. 7.37. Sectional view of swing check valve (Kemplay, 1980).

Ball and diaphragm valves are now the most widely used designs in fermenter and other biotechnology equipment (Leaver and Hambleton, 1992). Although ball valves are more robust, they contain crevices which make sterilization more difficult.

Check valves

The purpose of the check valve is to prevent accidental reversal of flow of liquid or gas in a pipe due to breakdown in some part of the equipment. There are three basic types of valve: swing check, lift check and combined stop and check with a number of variants. The swing check valve (Fig. 7.37) is most commonly used in fermenter designs. The functional part is a hinged disc which closes against a seat ring when the intended direction of flow is accidentally reversed.

Pressure-control valves

When planning the design of a plant for a specific process, the water, steam and air should be at different, but specified pressures and flow rates in different parts of the equipment. For this reason it is essential to control pressures precisely and this can be done using reduction or retaining valves.

PRESSURE-REDUCTION VALVES

Pressure-reduction valves are incorporated into pipelines when it is necessary to reduce from a higher to a lower pressure, and be able to maintain the lower

pressure in the downstream side within defined limits irrespective of changes in the inlet pressure or changes in demand for gas, steam or water.

PRESSURE-RETAINING VALVES

A pressure-retaining valve will maintain pressure in the pipeline upstream of itself, and the valve is designed to open with a rising upstream pressure. It is constructed with a reverse action of the pressure-reducing valve.

Safety valves

Safety valves must be incorporated into every air or steam line and vessel which is subject to pressure to ensure that the pressure will never exceed the safe upper limit recommended by the manufacturer or a code of practice and to satisfy government legislation and insurance companies. They must also be of the correct type and size to suit the operating conditions and be in sufficient numbers to protect the plant. The reliability of such valves is crucial.

In the simplest valves (Fig. 7.38), a spindle is lifted from its seating against the pressure of gas, steam or

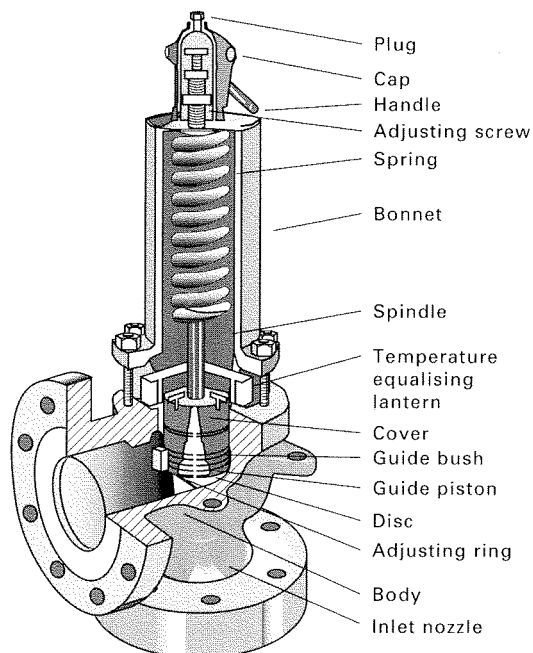


FIG. 7.38. Section of a safety valve (Kemplay, 1980).

liquid. Once the pressure falls below the value set by the tensioned spring, the spindle should return to its original position. However, the valve may stick open if waste material lodges on the valve seat and plant operators may interfere with the release pressure setting. Bursting/rupture discs may be used as an alternative and are of a more hygienic design than some valves (Leaver and Hambleton, 1992).

For GILSP operational categories, venting the escaping gas through the factory roof in emergencies would be satisfactory. At higher containment levels it would be necessary to treat the escaping gas. A kill tank with a HEPA venting filter is one possible solution, but not fully satisfactory.

It is also important to ensure that vessels do not collapse when vacuums occur. Vacuum release valves have been designed to cope with cold rinsing of a hot vessel or absorption of CO₂ when cleaning with caustic solutions (Maule, 1986). Shuttlewood (1984) has discussed design pressure and vacuum pressure in relation to vessel thickness requirements for design codes for American and British standards of safety.

Steam traps

In all steam lines it is essential to remove any steam condensate which accumulates in the piping to ensure optimum process conditions. This may be achieved by incorporating steam traps, which will collect and remove automatically any condensate at appropriate points in steam lines. A steam trap has two elements. One is a valve and seat assembly which provides an opening, which may be of variable size, to ensure

effective removal of any condensate. This opening may operate on an open/close basis so that the average discharge rate matches the steam condensation rate or the dimensions of this opening may be varied continually to provide a continuous flow of condensate. The second element is a device which will open or close the valve by measuring some parameter of the condensate reaching it to determine whether it should be discharged (Armer, 1991).

The steam trap may be designed to operate automatically on the basis of:

1. The density of the fluid by using a float (ball or bucket) which will float in water or sink in steam (Fig. 7.39).
2. By measuring the temperature of the fluid, closing the valve at or near steam temperature and opening it when the fluid has cooled to a temperature, say 8° below the saturated steam temperature of the steam. The sensing element is often a stainless-steel capsule filled with a water-alcohol mixture which expands when steam is present and presses a ball into a valve seat or contracts when cooler condensate is present and lifts the ball from the seating. These are used in thermostatic and balanced pressure steam traps (Figs 7.40 and 7.41).
3. By measuring the kinetic effects of the fluid in motion. At a given pressure drop, low-density steam will move at a much greater velocity than higher density condensate. The conversion of pressure energy into kinetic energy can be used to control the degree of opening of a valve. This type of steam trap is not used very widely.

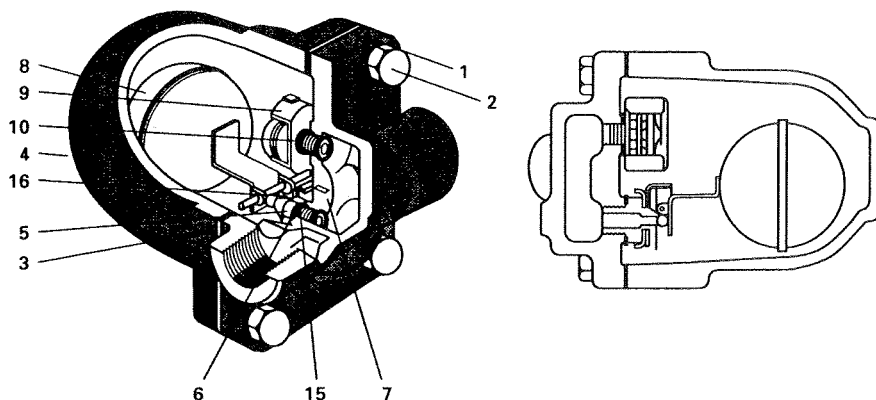


FIG. 7.39. Ball float steam trap with thermostatic air vent (Spirax/Sarco, Cheltenham, England). (5) valve seat; (8) ball float and lever; (9) air vent.

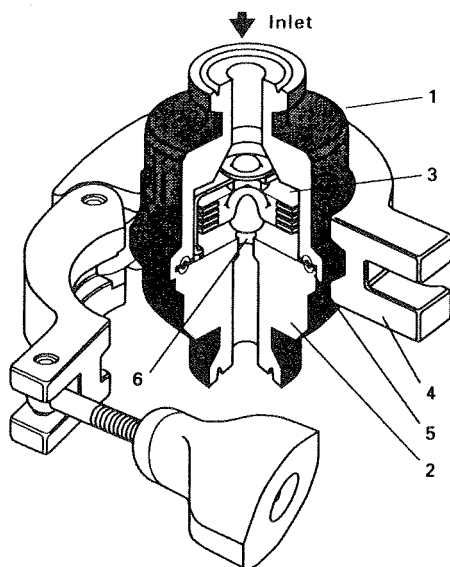


FIG. 7.40. Balanced pressure steam trap (Spirax/Sarco, Cheltenham, England). (1) body; (2) end connection; (3) element; (4) Tri-Clover clamp; (5) Tri-Clover joint gasket; (6) ball seat.

Armer (1991) thinks that balanced pressure thermostatic traps are the most suitable for autoclaves and sterilizers. Some can operate close to steam-saturation temperature with little back up of condensate. Sarco Ltd (Cheltenham, U.K.) make a hermetically sealed balanced pressure steam trap (Model No. SBP 30; Fig. 7.42) which has been used with a high containment level pilot-plant fermenter (Hambleton *et al.*, 1991).

Any steam condensate may be (a) returned to the

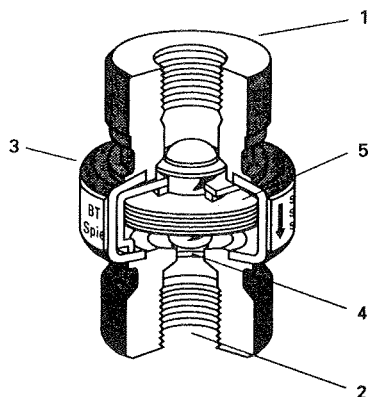


FIG. 7.41. Stainless steel thermostatic steel trap (Spirax/Sarco, Cheltenham, England). (1) inlet; (2) outlet; (3) body; (4) ball seat; (5) element.

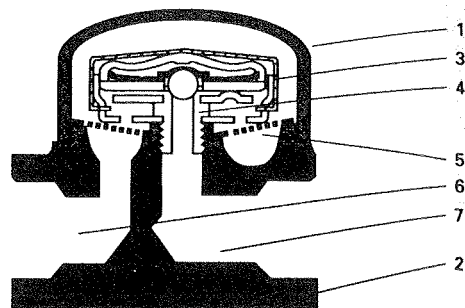


FIG. 7.42. SBP 30 Sealed balanced pressure thermostatic steam trap (Spirax/Sarco, Cheltenham, England). (1) cover; (2) body; (3) element/capsule; (4) ball seat assembly; (5) strainer screen; (6) inlet; (7) outlet.

boiler, (b) used in a steam condensate line, (c) vented to waste or (d) piped to a kill tank in a containment location

Complete loss of contents from a fermenter

Leaver and Hambleton (1991) have suggested building a wall or dyke around a fermenter which would retain the entire contents were they to leak out. The escaping fluid would then be pumped into a suitable vessel for sterilization. The floor should be covered with an impervious epoxy based surface and coved up the walls (Kennedy *et al.*, 1990).

Testing new fermenters

When a new fermenter (10 dm³ or larger) has finally been assembled in a factory it must be hydraulically pressure tested and checked by independent inspectors using nationally approved test procedures. A vessel which has passed the approved tests is given a certificate which is recognized by insurance companies which will allow it to be operated in a laboratory or factory. If any subsequent modifications are made to a vessel in this size range it must be retested and certificated before it can be legally used. Thus a number of fermenter manufacturers cut extra 'O' ring grooves and steam traces in head plates or ports to satisfy higher containment-level needs so that the vessels will not have to be modified subsequently and to avoid extra pressure checks.

OTHER FERMENTATION VESSELS

Some of the other forms of fermentation vessels will now be considered. These vessels have more limited applications and have been developed for specific purposes or closely related processes. Some are historical developments, such as the earliest forms of packed tower, others were being developed in parallel with the standard mechanically stirred fermenter during the 1940s, while other approaches are more recent. Aspects of this topic have been reviewed by Prokop and Votruba (1976), Katinger (1977), Hamer (1979), Levi *et al.* (1979), Solomons (1980), Schugerl (1982, 1985), Sittig (1982), Winkler (1990) and Atkinson and Mavituna (1991a).

The Waldhof-type fermenter

The investigations on yeast growth in sulphite waste liquor in Germany, Japan and the United States of America led to the development of the Waldhof-type fermenter (Inskeep *et al.*, 1951; Watanabe, 1976). Inskeep *et al.* (1951) have given a description of a production vessel based on a modification of the original design of Zellstoffabrik Waldhof. The fermenter was of carbon steel, clad in stainless steel, 7.9 m in diameter and 4.3-m high with a centre draught tube 1.2 m in diameter. A draught tube was held by tie rods attached to the fermenter walls. The operating volume was 225,000 dm³ of emulsion (broth and air) or 100,000 dm³ of broth without air. Non-sterile air was introduced into the fermenter through a rotating pin-wheel type of aerator, composed of open-ended tubes rotating at 300 rpm (Fig. 7.43). The broth passed down the draught tube from the outer compartment and reduced the foaming.

Acetators and cavitators

Fundamental studies by Hromatka and Ebner (1949) on vinegar production showed that if *Acetobacter* cells were to remain active in a stirred aerated fermenter, the distribution of air had to be almost perfect within the entire contents of the vessel. They solved the full-scale problem by the use of a self-aspirating rotor (Ebner *et al.*, 1967). In this design (Fig. 7.44), the turning rotor sucked in air and broth and dispersed the mixture through the rotating stator (d). The aerator also worked without a compressor and was self-priming.

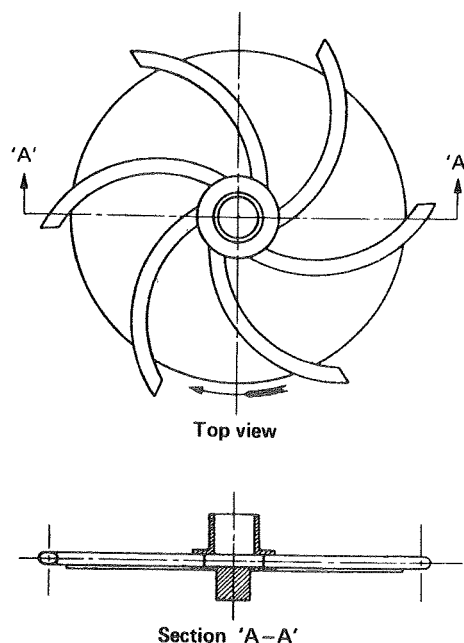


FIG. 7.43. Top view and section of a Waldhof aeration wheel (Inskeep *et al.*, 1951).

Vinegar fermentations often foam and chemical antifoams were not thought feasible because they would decrease aeration efficiency (Chapter 9) and additives were not desirable in vinegar. A mechanical defoamer therefore had to be incorporated into the vessel and as foam builds up it is forced into a chamber in which a rotor turns at 1,000 to 1,450 rpm. The centrifugal force breaks the foam and separates it into gas and liquid. The liquid is pumped back into the fermenter and the gas escapes by a venting mechanism. Descriptions of the design and various sizes of model have been given by Ebner *et al.* (1967). Fermenters of this design are manufactured by Heindrich Frings, Bonn, Germany. An illustration of the basic components is given in Fig. 7.45. In 1981, 440 acetators were in operation all over the world with a total production of 767×10^6 dm³ year⁻¹. The major vinegar producers were the U.S.A. (152×10^6 dm³), France (90×10^6 dm³) and Japan (46×10^6 dm³) while the remainder was produced by over 50 countries (Ebner and Follmann, 1983).

Chemap AG of Switzerland manufacture the Vinegator. A self-aspirating stirrer and a central suction tube aerates a good recirculation of liquid. Additional air is provided by a compressor. Foam is broken down by a mechanical defoamer (Ebner and Follmann, 1983).

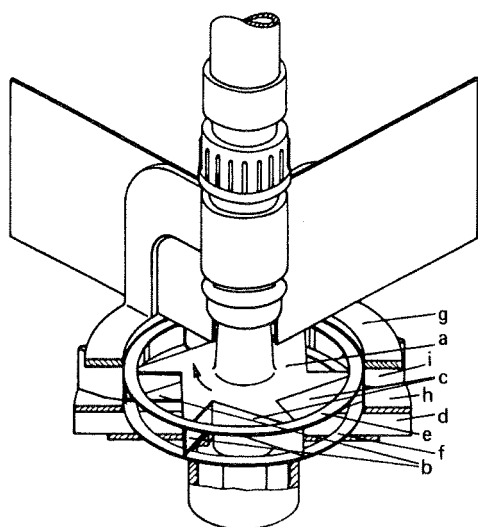


FIG. 7.44. Axonometric view of the self-priming aerator used with the Frings generator (Ebner *et al.*, 1967). The turbine is designed as a hollow body (a) with openings which are arranged radially and open against the direction of rotation (b). The openings are shielded by vertical sheets (c). The turbine sucks liquid from above and below and mixes it with air sucked in through the openings. The suspension is thrown through the stator (d) towards the circumference of the tank. An upper and lower ring on the turbine (e, f) helps to direct and regulate the air-liquid suspension. The stator (d) consists of an upper and lower ring (g, h) which are connected by vertical sheets (i) inclined at about 30° towards the radius.

At least three other vinegar fermenters are no longer manufactured. The Bourgeois process was sold in Europe between 1955 and 1980 and the Fardon process between 1960 and 1975. The Yeomans cavitator was sold in the U.S.A. between 1959 and 1970 (Cohee and Steffen, 1959; Mayer, 1961; Ebner and Follmann, 1983). The fermenter had an agitator of different design, but similar operating principles to the acetator. Uniform distribution of air bubbles was obtained by means of the circulation pattern created by the centrally located draught (draft) tube. The agitator withdrew liquid from the draught tube and pushed liquid into the main part of the vessel. The outer level rose and overflow occurred back into the top of the draught tube.

The tower fermenter

It is difficult to formulate a single definition which encompasses all the types of tower fermenter. Their main common feature appears to be their height:diameter ratio or aspect ratio. Such a definition has been

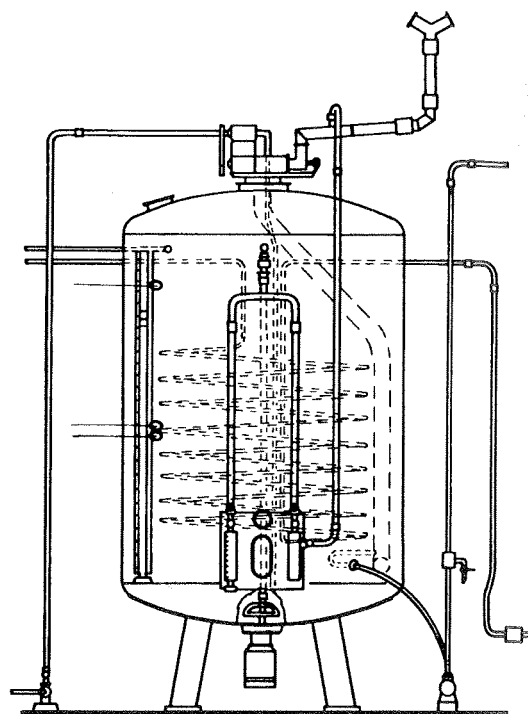


FIG. 7.45. Diagram of a section through a Frings generator fermenter used for the manufacture of vinegar. The fermenter, which can be used semicontinuously or continuously, employs vortex stirring (Greenshields, 1978).

given by Greenshields *et al.* (1971) who described a tower fermenter as an elongated non-mechanically stirred fermenter with an aspect ratio of at least 6:1 for the tubular section or 10:1 overall, through which there is a unidirectional flow of gases. Several different types of tower fermenter exist and these will be examined in broad groups based on their design.

The simplest types of fermenter are those that consist of a tube which is air sparged at the base (bubble columns). This type of fermenter was first described for citric acid production on a laboratory scale (Snell and Schweiger, 1949). This batch fermenter was in the form of a glass column having a height:diameter ratio of 16:1 with a volume of 3 dm³. Humid sterile air was supplied through a sinter at the base. Steel *et al.* (1955) reported an increase in scale to 36 dm³ for a fermenter of this type. Pfizer Ltd has always used non-agitated tower vessels for a range of mycelial fermentation processes including citric acid and tetracyclines (Solomons, 1980; Carrington *et al.*, 1992). Recently Pfizer Ltd sold their citric acid interests to Arthur Daniels Midland who are operating such vessels up to 23 m high (Burnett, 1993).

Volumes of between 200 m³ and 950 m³ have been reported elsewhere (Rohr *et al.*, 1983).

In 1965 the brewing industry began to use tower fermenters which were more complex in design and could be operated continuously. Hall and Howard (1965) described small-scale fermenters that consisted of water jacketed tubes of various dimensions which were inclined at angles of 9 to 90° to the horizontal. Air and mash were passed in at the base and effluent beer was removed at the top.

A vertical-tower beer fermenter design (Chapter 2) was patented by Shore *et al.* (1964). Perforated plates were positioned at intervals in the tower to maintain maximum yeast production. The settling zone which could be of various designs, was to provide a zone free of rising gas so that the cells could settle and return to the main body of the tower and the clear beer could be removed. This design must be considered as an intermediate between single- and multistage systems. Towers of up to 20,000 dm³ capacity and capable of producing up to 90,000 dm³ day⁻¹ have been installed. Greenshields and Smith (1971) commented that it was difficult to predict the upper operating limits for these fermenters. Experiments with particular yeast strains in pilot-size towers were essential to establish optimum full-scale operating conditions.

The next group of tower fermenters are the multistage systems, first described by Owen (1948) and Victorero (1948) for brewing beer, although these systems were not used on an industrial scale. Later work reported using these systems includes continuous cultures of *E. coli* (Kitai *et al.*, 1969), bakers' yeast (Prokop *et al.*, 1969) and activated sludge (Lee *et al.*, 1971; Besik, 1973). The fermenters used by all these workers were basically similar. Each consisted of a column forming the body of the vessel and a number of perforated plates which were positioned across the fermenter, dividing it into compartments. Approximately 10% of the horizontal plate area was perforated. The possibility of introducing media into individual stages independently was discussed by Lee *et al.* (1971). Besik (1973) described a down-flow tower in which substrate was fed in at the top and overflowed through down spouts to the next section while air was supplied from the base. Schugerl, Lucke and Oels (1977) have written a comprehensive general review.

Cylindro-conical vessels

The use of cylindro-conical vessels in the brewing of lager was first proposed by Nathan (1930), but his ideas

were not adopted for the brewing of lagers and beers until the 1960s (Hoggan, 1977). Breweries throughout the world have now adopted this method of brewing. The vessel (Fig. 7.46) consists of a stainless-steel vertical tube with a hemispherical top and a conical base with an included angle of approximately 70° (Boulton, 1991).

Aspect ratios are usually 3:1 and fermenter heights are 10 to 20 m. Operating volumes are chosen to suit the individual brewery requirements, but are often 150,000 to 200,000 dm³. Vessels are not normally agitated unless a particularly flocculant yeast is used, but small impellers may be used to ensure homogeneity when filling with wort (Boulton, 1991). In the vessel, the wort is pitched (inoculated) with yeast and the fermentation proceeds for 40 to 48 hours. Mixing is achieved by the generation of carbon dioxide bubbles that rise rapidly in the vessel. Temperature control is monitored by probes positioned at suitable points within the vessel. A number of cooling jackets are fitted to the vessel wall to regulate and cause flocculation and settling of the yeast (Ulenberg *et al.*, 1972; Maule, 1986;

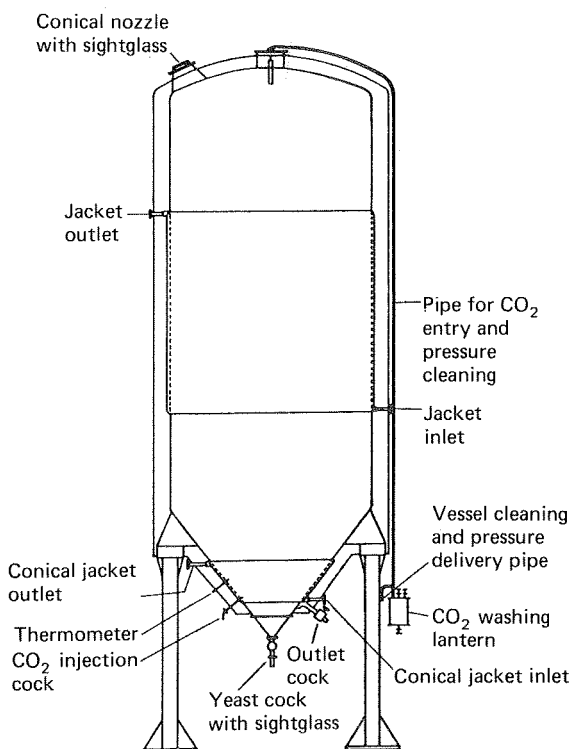


FIG. 7.46. Cylindro-conical fermentation vessel (Hough *et al.*, 1971).

Boulton, 1991). The fermentation is terminated by the circulation of chilled water via the cooling jackets which results in yeast flocculation. Thus, it is necessary to select a yeast strain which will flocculate readily in the period of chilling. Part of this yeast may be withdrawn and used for repitching another vessel. The partially cleared beer may be left to allow a secondary fermentation and conditioning. Some of the advantages of this vessel in brewing are:

1. Reduced process times may be achieved due to increased movement within the vessel.
2. Primary fermentation and conditioning may be carried out in the same vessel.
3. The sedimented yeast may be easily removed since yeast separation is good.
4. The maturing time may be reduced by gas washing with carbon dioxide.

Air-lift fermenters

An air-lift fermenter (Fig. 7.47) is essentially a gas-tight baffled riser tube (liquid ascending) connected to a downcomer tube (liquid descending). Figure 7.47a shows an external riser and Fig. 7.47b an internal riser. Air or gas mixtures are introduced into the base of the riser by a sparger during normal operating conditions. The driving force for circulation of medium in the vessel is produced by the difference in density between the liquid column in the riser (excess air bubbles in the medium) and the liquid column in the downcomer (depleted in air bubbles after release at the top of the loop). Circulation times in loops of 45-m height may be 120 seconds. More details on liquid circulation and mixing characteristics are discussed by Chen (1990). This type of vessel can be used for continuous culture. The first patent for this vessel was obtained by Scholler and Seidel (1940).

It would be uneconomical to use a mechanically stirred fermenter to produce SCP (single-cell protein) from methanol as a carbon substrate, as heat removal would be needed in external cooling loops because of the high rate of aeration and agitation required to operate the process. To overcome these problems, particularly that of cooling the medium when mechanical agitation is used, air-lift fermenters with outer or inner loops (Fig. 7.47) were chosen. Development work for operational processes for SCP has been done by ICI plc in Great Britain (Taylor and Senior, 1978; Smith, 1980), Hoechst AG-Uhde GmbH in Germany (Faust *et*

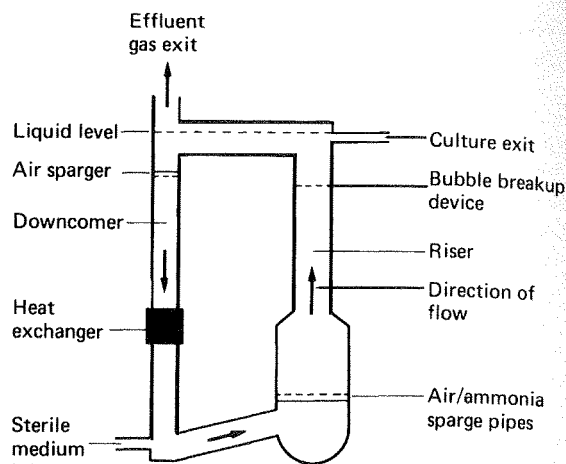


Fig. 7.47a. Air-lift fermenter with outer loop (Taylor and Senior, 1978).

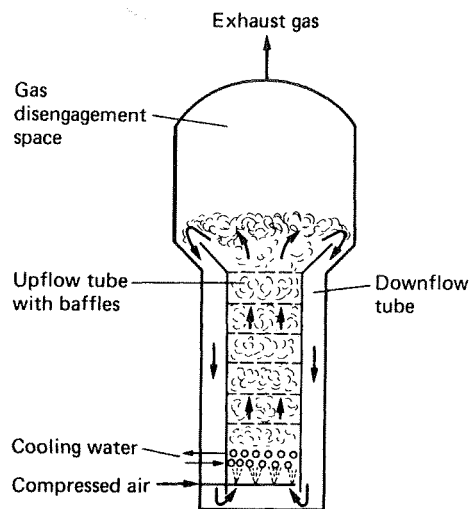


Fig. 7.47b. Air-lift fermenter with inner loop (Smith, 1980).

al., 1977) and Mitsubishi Gas Chemical Co. Inc. in Japan (Kuraishi *et al.*, 1978). Although ICI plc initially used an outer-loop system in their pilot plant, all three companies preferred an inner-loop design for large-scale operation. Hamer (1979) and Sharp (1989) have reviewed these fermenters. In the ICI plc continuous process, air and gaseous ammonia were introduced at the base of the fermenter. Sterilized methanol, other nutrients and recycled spent medium were also introduced into the downcomer. Heat from this exothermic fermentation was removed by surrounding part of the downcomer with a cooling jacket in the pilot plant,

while at full scale ($2.3 \times 10^6 \text{ dm}^3$) it was found necessary to insert cooling coils at the base of the riser.

Unfortunately the production of SCP for animal feed has not proved an economic proposition because of the price of methanol and the competition from animal feeds based on arable protein crops. ICI plc's vessel at Billingham, U.K. has now been dismantled.

In 1964, Rank Hovis McDougall decided to develop a protein-rich food primarily for human consumption

(Trinci, 1992). They have grown *Fusarium graminearum* on a wheat starch based medium using a modified ICI plc 40 m^3 air-lift fermenter (Fig. 7.48) to produce the myco-protein Quorn. The use of an air-lift fermenter for culture of a mycelial fungus would seem unusual as lower rates of oxygen transfer occur in a viscous culture which give rise to lower biomass yields. Because low shear conditions are present in the vessel, long fungal hyphae can be cultured (the preferred product

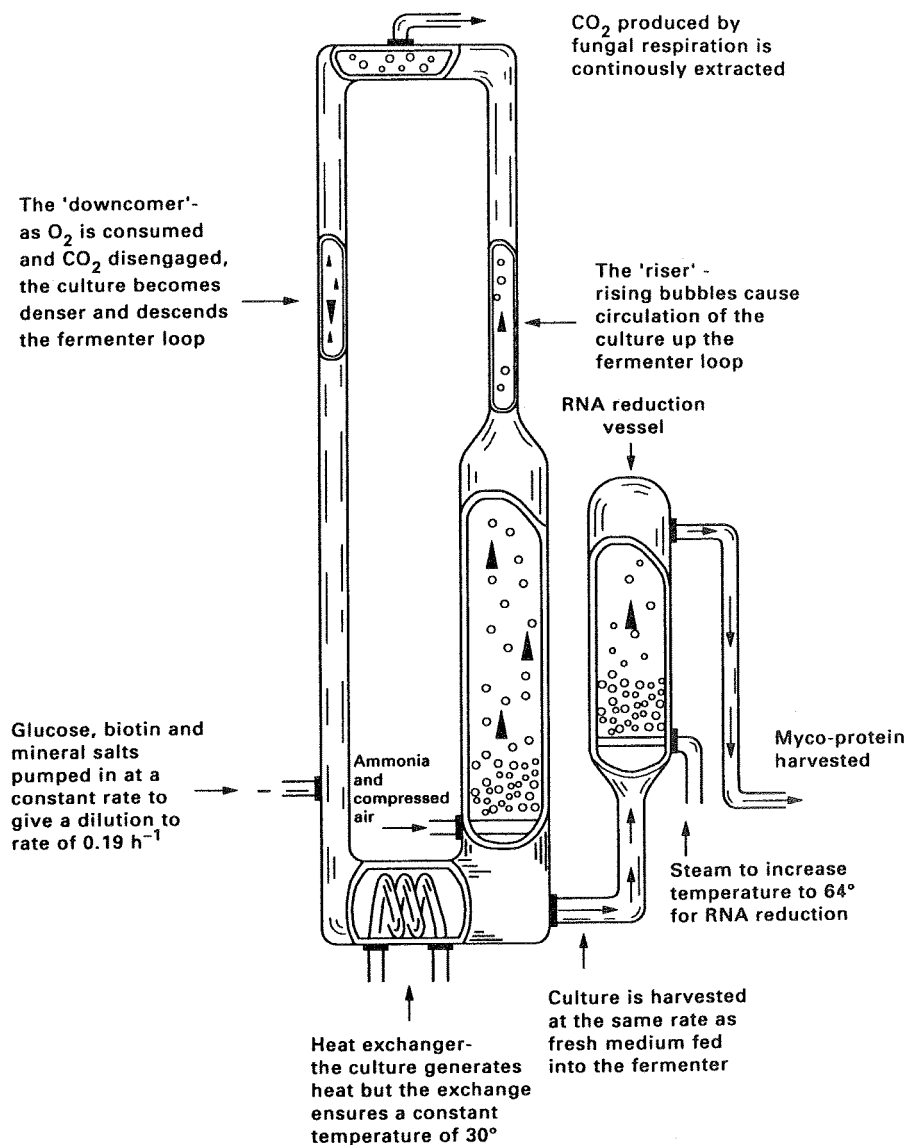


FIG. 7.48. Schematic diagram of the air-lift fermenter used by Marlow Foods at Billingham, England, for the production of myco-protein in continuous flow culture (Trinci, 1992).

form) even though production yields are only 20 g dm^{-3} . At the present time a fermenter to produce 10,000 tonnes per annum of myco-protein is considered economically feasible.

Okabe *et al.* (1993) modified a 3-dm^3 air-lift fermenter by putting stainless steel four-mesh sieves at the top and bottom of the draught tube to manipulate the morphology of *Aspergillus terreus* for optimum production of itaconic acid as the culture circulates in the vessel flow path. The fungal morphology was an intermediate state between pellets and pulp. Using this vessel, the itaconic acid production rate ($\text{g dm}^{-1} \text{h}^{-1}$) was double that obtained with a stirred fermenter or an air-lift fermenter with a conventional draught tube.

Work has also begun to examine oxygen transfer rates with modified draught tubes. Carrington *et al.* (1992) used a 20 m^3 pilot-scale bubble column fermenter fitted with an internal helical cooling coil (Fig. 7.49) or a solid draught tube. The fermentation studied was a commercial *Streptomyces* antibiotic fermentation in a complex medium which produced a viscous non-Newtonian broth. Tracer studies indicated that the vessel fitted with only the cooling coil behaved like an air-lift fermenter with a region of good mixing in the zone above the cooling coil. The coil acted as a leaky draught tube with back mixing taking place between the coils into the riser section. No poorly oxygenated zones were observed. Liquid velocities of 1 m sec^{-1} were measured giving circulation times of 9 to 12 seconds and mixing times of 14 to 18 seconds. The K_La at different power inputs and viscosities was found to increase almost linearly with increasing power input and decreased exponentially with increasing viscosity. When a solid draught tube was installed inside the cooling coil the circulation time was similar, but the mixing time increased to 18 to 24 seconds. The K_La was also determined at different power inputs and viscosities. This gave a reduction of 5 to 25% in K_La with the biggest reduction at a high viscosity.

Wu and Wu (1990) compared K_La s in a range of mesh draught tubes and a solid draught tube in an air-lift vessel of 15 dm^3 working volume. When a 24-mesh tube was used at high superficial gas velocities the K_La was double that of the same vessel with a solid draught tube.

Bakker *et al.* (1993) have developed a multiple air-lift fermenter in which three air-lift fermenters with internal loops are incorporated into one vessel (Fig. 7.50). Fresh medium is fed into the central compartment, depleted medium overflows into the middle compartment, from here to the outer compartment where medium is eventually discharged. The hydrodynamics

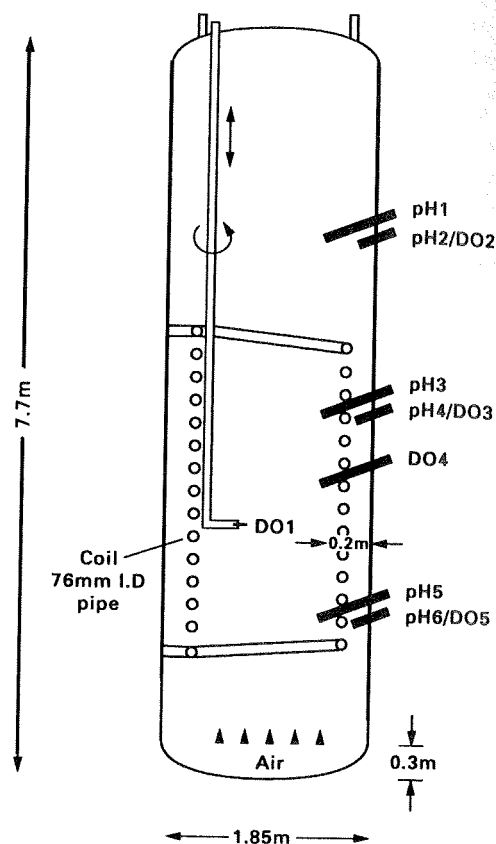


Fig. 7.49. Schematic diagram of 20 m^3 bubble column fermenter fitted with internal cooling coil (Carrington *et al.*, 1992). (pH and DO indicate positions of pH and oxygen electrodes.)

and mixing in the middle compartment were found to be comparable with those obtained with conventional internal loop air-lift vessels, although some tangential liquid flow was observed in this compartment.

The deep-jet fermenter

Some designs of continuous culture fermenter achieve the necessary mechanical power input with a pump to circulate the liquid medium from the fermenter through a gas entrainer and back to the fermenter (Fig. 7.51; Hamer, 1979; Meyrath and Bayer, 1979). Two basic construction principles have been used for the gas entrainer nozzles. The injector and the ejector (Fig. 7.52; Reuss, 1992). In an injector a jet of medium is surrounded by a jet of compressed air. The gas from the outlet enters the larger tube with a nozzle velocity of 5 to 100 m s^{-1} and expands in the tube to

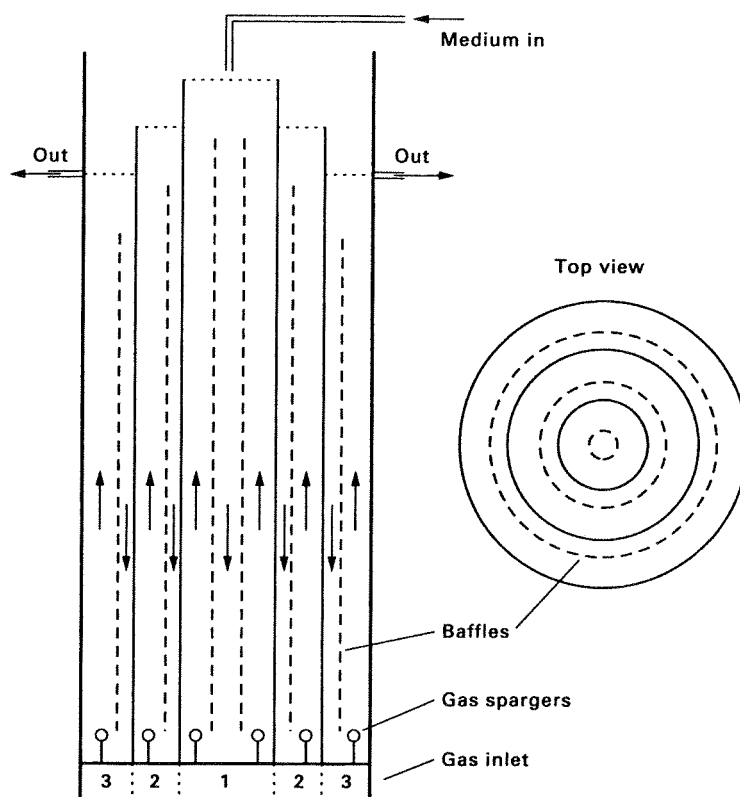


FIG. 7.50. Three-compartment multiple air-lift loop fermenter (Bakker *et al.*, 1993). Cross-sectional side and top view.

form large air bubbles which are dispersed by the shear of the water jet. In an ejector the liquid jet enters into a larger converging-diverging nozzle and entrains the gas around the jet. The gas which is sucked into the converging-diverging jet is dispersed in that zone. One of the industrial-scale fermenters using the ejector design is marketed by Vogelbusch (Vogelbusch AG, Vienna, Austria). Partially aerated medium is pumped by a multiphase pump through a broth cooler to an air entrainer above the fermenter. The air-medium mixture falls down a slightly conical shaft at a high velocity and creates a turbulence in the fermenter. Two-thirds of the exhaust gas is vented from the fermenter headspace and the remainder via the multiphase pump. Oxygen-transfer rates of $4.5 \text{ g dm}^{-3} \text{ h}^{-1}$ with an energy consumption of $1 \text{ kW h}^{-1} \text{ kg}^{-1}$ have been achieved for industrial-scale yeast production from whey using such a fermentation system.

The cyclone column

Dawson (1974) developed the cyclone column, par-

ticularly for the growth of filamentous cultures (Fig. 7.53). The culture liquid was pumped from the bottom to the top of the cyclone column through a closed loop. The descending liquid ran down the walls of the column in a relatively thin film. Nutrients and air were fed in near the base of the column whilst the exhaust gases left at the top of the column. Good gas exchange, lack of foaming and limited wall growth have been claimed with this fermenter. Dawson (1974, 1988) has listed a number of potential bacterial, fungal and yeast applications including the batch production of a vaccine for scours in calves with the vessel being operated as batch, continuous or fed-batch.

The packed tower

The packed tower is a well established application of immobilized cells. A vertical cylindrical column is packed with pieces of some relatively inert material, e.g. wood shavings, twigs, coke, an aggregate or polythene. Initially both medium and cells are fed into the top of the packed bed. Once the cells have adhered to

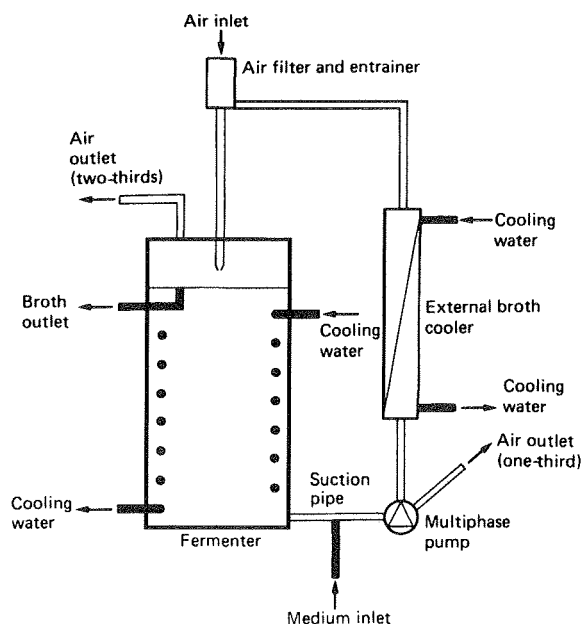


FIG. 7.51. Diagram of the Vogelbusch deep-jet fermenter system (from Schreier, 1975; Hamer, 1979).

the support and are growing well as a thin film, fresh medium is added at the top of the column and the fermented medium is removed from the bottom of the column. The best known example is the vinegar generator, in which ethanol was oxidized to acetic acid by strains of *Acetobacter* supported on beech shavings; the first recorded use was in 1670 (Mitchell, 1926). More recently, packed towers have been used for

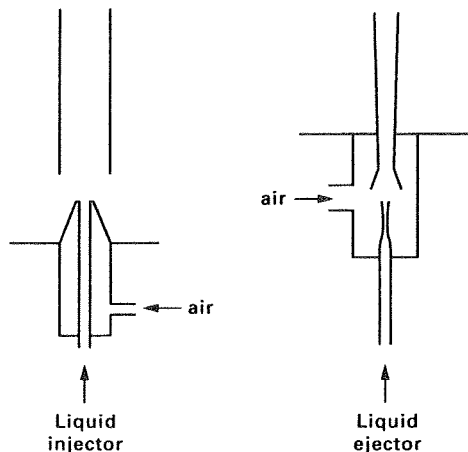


FIG. 7.52. Gas entrainer nozzles of deep-jet fermenters (Reuss, 1992).

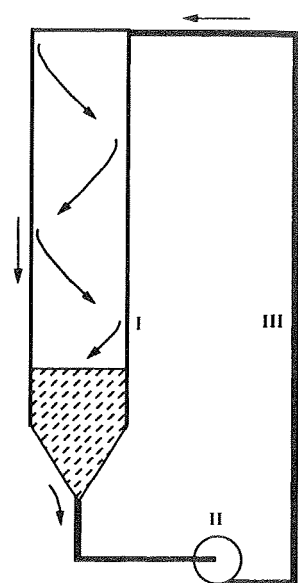


FIG. 7.53. Schematic diagram of a cyclone column fermenter. (I) cyclone column; (II) circulating pump; (III) recirculating limb (Dawson, 1974).

sewage and effluent treatment (Noble *et al.*, 1964). In treatment of gas liquor, a column was packed to a height 7.9 m with 'Dowpac', a polystyrene derivative. The main advantages compared with other methods of effluent treatment being its simplicity of operation and a saving in land because of the increased surface areas within the column. Other possible applications with immobilized cells are now being investigated.

Rotating-disc fermenters

Rotating-disc contactors have been used in effluent treatment (Chapter 11). They utilize a growing microbial film on slow rotating discs to oxidize the effluent. Anderson and Blain (1980) have used the same principle to construct small fermenters of up to 40-dm³ working volume. A range of filamentous fungi, including species of *Aspergillus*, *Rhizopus*, *Mucor* and *Penicillium*, could be grown on the polypropylene discs. It has been possible to obtain yields of 80 g dm⁻³ of citric acid from *A. niger* using this design of fermenter.

ANIMAL CELL CULTURE

Interest in the *in vitro* cultivation of animal cells has developed because of the need for large scale produc-

tion of monoclonal antibodies, hormones, vaccines and other products which are difficult or impossible to produce synthetically or by using other culture techniques.

Animal cells are usually more nutritionally demanding than microbial cells. They lack cell walls which makes them sensitive to shear and extremes of osmolarity. The doubling times are normally 12 to 48 hours and cell densities in suspension cultures rarely exceed 10^6 to 10^7 cells cm^{-3} . Two distinct modes of growth can be recognized:

1. *Anchorage dependent cells.* These cells require a solid support for their replication. They produce cellular protrusions (pseudopodia) which allow them to adhere to positively charged surfaces and often grow as monolayers.
2. *Anchorage independent cells.* These cells do not require a support and can grow as a suspension in submerged culture. Established and transformed cell lines are normally in this category.

There are also intermediate categories of cells which may be grown as anchorage dependent or suspension cells. It is also possible to grow some anchorage dependent cells in suspension, provided the cells can be grown on suitable microcarriers.

A range of free and immobilized culture systems have therefore been developed for the culture of different lines of animal cells and their products. The range of available equipment and techniques may be confusing to scientists not familiar with animal cells who may have to decide on the most appropriate culture systems for laboratory, pilot or production scale. Because of the relatively small quantities of product required, the volume at production scale may only be 100 to 10,000 dm^3 . Some useful introductions include Griffiths (1986, 1988), Propst *et al.* (1989), Lavery (1990) and Bliem *et al.* (1991).

Shear is a phenomenon recognized as being critical to the scale up of animal cell culture processes, irrespective of the cell line or reactor configuration (Bliem and Katinger, 1988). Shear may influence the cell culture causing damage which may result in cell death or metabolic changes. The shear sensitivity of animal cells may vary between cell lines, the phase of growth or with a change to fresh medium.

Mijnbeek (1991) has reviewed research on shear stress of free and immobilized animal cells in stirred fermenters and air-lift fermenters. It was concluded that the predominant damage mechanism in both types

of vessel was due to sparging and the break up of bubbles on the medium surface. This type of damage causing cell death might be reduced by increasing the height to diameter ratio in the vessel, increasing the bubble size, decreasing the gas flow rate or by adding protective agents. Other damage mechanisms in stirred fermenters are caused by cell-microcarrier eddies and microcarrier-microcarrier interactions. Damage of this type may be reduced by reducing the impeller speed, impeller diameter, microcarrier size and concentration or by increasing the viscosity of the medium.

The maximum cell densities obtainable in stirred and air lift fermenters are often only 10^6 cells cm^{-3} . This is not ideal if secreted proteins are present only in very low concentrations and mixed with other proteins present in the original medium. A number of modified fermenters and reactors have been developed to grow cells at higher concentrations using microcarriers, encapsulation, perfusion, glass beads or hollow fibres to obtain the required product at higher concentrations.

Stirred fermenters

Unmodified stirred fermenters have been used for the batch production of some virus vaccines (Propst *et al.*, 1989), but modified vessels are used for most cultures (Propst *et al.*, 1989; Lavery, 1990). The modifications made to fermenters are to reduce the possibility of cell damage due to shear, heat or contamination. Marine propellers revolving at a slow speed (10 to 100 rpm) will normally provide adequate mixing. Hemispherical bottoms on the vessels will ensure better mixing of the broth at slow stirrer speeds. Water jacket heating is often preferred since heating probes may give rise to localized zones of high temperature which might damage some of the cells. Magnetic driven stirrers may be used to reduce the risk of contamination. A novel sparger-impeller design which improves aeration at slow speeds is incorporated into the Celligen system manufactured by New Brunswick, U.S.A. (Fig. 7.54; Beck *et al.*, 1987). When the impeller rotates the swept-back ports produce a negative pressure inside the hollow impeller complex. This creates a suction lift that produces highly efficient circulation and gas transfer at low rpm without damaging the cells. Gases are introduced through a ring sparger into the medium as it circulates through a fine stainless steel mesh jacket which excludes cells and microcarriers. Because the gas sparging is restricted to a relatively small zone, foaming

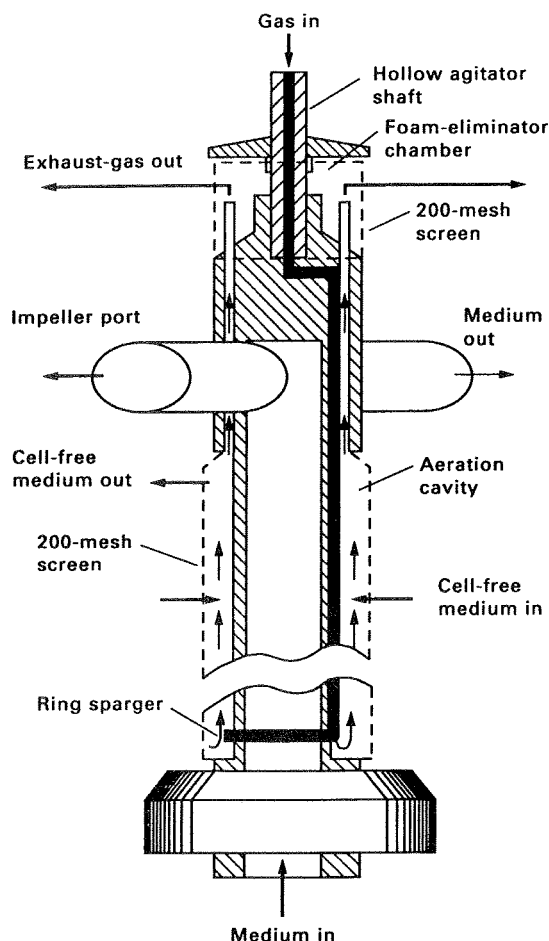


FIG. 7.54. Sparger-impeller in the Celligen cell-culture fermenter (New Brunswick Ltd, Hatfield, England).

is reduced, and any foam formed is broken up by the mesh in the foam eliminator chamber.

Air-lift fermenters

Air-lift fermenters have proved ideal for growth of some cell lines because of the gentle mixing action and reduced shear forces when compared with those in stirred vessels. The absence of a stirrer and associated seals excludes a potential source of contamination (Griffiths, 1988; Propst *et al.*, 1989; Lavery, 1990).

Vessels of 1000 to 2000 dm³ are commercially available. Celltech Ltd, U.K., have used such vessels to produce monoclonal antibodies from hybridoma cells (Wilkinson, 1987).

Microcarriers

Microcarriers may provide a solution to the problem of growing anchorage-dependent cultures in suspension culture in fermenters, by providing the necessary surface for attachment. Animal cells normally have a net negative surface charge and will attach to a positively charged surface by electrostatic forces. Van Wezel (1967) made use of this property and attached anchorage dependent cells to chromatographic grade DEAE Sephadex A-50 resin beads and suspended the coated beads in a slowly stirred liquid medium. The density of the electrostatic charges on the microcarrier surface is critical if cell growth at high bead concentrations is to be achieved. If the net charge density on the bead surface is too low, cell attachment will be restricted. When the charge density is too high, apparent toxic effects will limit cell growth (Fleischaker, 1987). A number of microcarrier beads, manufactured from dextran, cellulose, gelatin, plastic or glass, are now commercially available (Fleischaker, 1987; Butler, 1988). Dextran microcarriers have been used for large scale production of viral vaccines and interferon. Unfortunately some of the microcarriers cost £1200 to £1500 kg⁻¹.

Encapsulation

At least three methods of encapsulation have been developed (Griffiths, 1988; Lavery, 1990). Encapsel, a technique developed by Damon Biotechnology, U.S.A., traps the animal cells in sodium alginate spheres which are then coated with polylysine to form a semi-permeable membrane. The enclosed alginate gel is solubilized with sodium citrate to release the cells into free suspension within the capsules which are usually 50 to 500- μ m diameter. After a few weeks growth it is possible to obtain cell concentrations of 5×10^8 cm⁻³ and product levels 100 times higher than with free cells can be achieved. The high molecular weight products are retained within the capsules. This technique has been used commercially for monoclonal antibody production.

In a second method the cells are entrapped in calcium alginate which will allow high molecular weight products to diffuse into the medium. Unfortunately, the spheres tend to be 0.5 to 1.0 mm diameter and slow diffusion into the spheres may cause nutrient limitations. Alternatively the cells can be entrapped in agarose beads in which the cells are contained in a

honeycomb matrix within the gel. These capsules have a wide size distribution and a low mechanical strength compared with alginate ones.

Hollow fibre chambers

Anchorage dependent cells can be cultured at densities of 10^8 cells cm^{-3} using bundles of hollow fibres held together in cartridge chambers (Hirschel and Gruenberg, 1987; Griffiths, 1988). The cells are grown in the extra capillary spaces (ECS) within the cartridge. Medium and gases diffuse through from the capillary lumen to the ECS. The molecular weight cut-off of the fibre walls may be selected so that the product is retained in the ECS or released into the perfusing medium. Many chambers will be needed for scale-up because of diffusion limitations in larger chambers. These have been used to study production of monoclonal antibodies, viruses, gonadotropin, insulin and antigens.

Packed glass bead reactors

Packed glass bead reactors have proved to be useful for long term culture of attached dependent cell lines. It is possible to obtain cell densities of 10^{10} viable cells in a 1-dm^3 vessel with moderate medium flow rates through the vessel (Fig. 7.55; Propst *et al.*, 1989). Increasing the size of vessels causes problems with mass transfer of oxygen and nutrients and scale up can be

achieved by increasing the number of small vessels. This technique is available only as a contract production service from Bioresponse Inc. of California, U.S.A.

Perfusion cultures

Perfusion culture is a technique where modified fermenters of up to 100 dm^3 are gently stirred and broth is withdrawn continuously from the vessel and passed through a stainless steel or ceramic filter. This type of culture is sometimes referred to as spin culture, since the filter is spun to prevent blocking with cells. The filtered medium is pumped to a product reservoir and fresh medium is pumped into the culture vessel. The rate of addition and removal can be regulated depending on the cell concentration in the vessel. With this method it is possible to obtain cell densities 10 to 30 times higher than the maximum cell density in an unmodified fermenter (Lydersen, 1987; Tolbert *et al.*, 1988; Lavery, 1990). This procedure has been used commercially by Invitron Corp. U.S.A. This company has patented a gentle 'sail' agitator to prevent cell damage in a 100-dm^3 vessel. Attachment dependent cells have been grown on microcarriers using sail agitators rotating at 8 to 12 rpm.

At Hybridtech Inc. USA, animal cells have been immobilized on a ceramic matrix and medium is perfused through this matrix. In this way high cell densities can be maintained. The apparatus is marketed as the 'Opticell' (Lydersen, 1987).

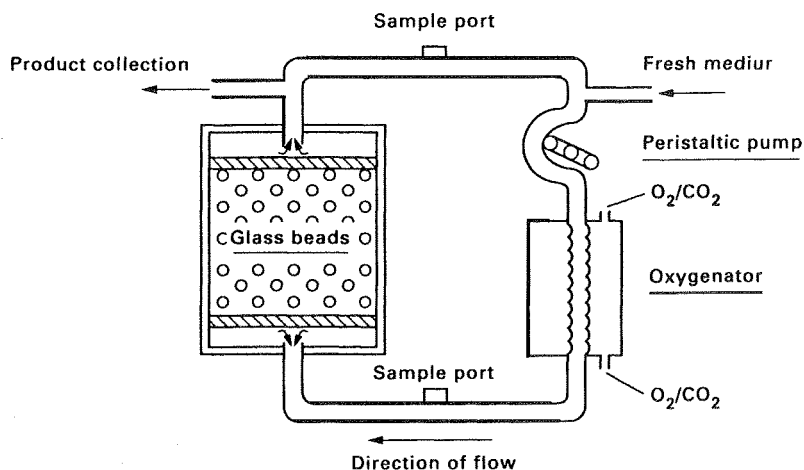


FIG. 7.55. Schematic diagram of a glass bead reactor (Browne *et al.*, 1988).

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Instrumentation and Control

INTRODUCTION

THE SUCCESS of a fermentation depends upon the existence of defined environmental conditions for biomass and product formation. To achieve this goal it is important to understand what is happening to a fermentation process and how to control it to obtain optimal operating conditions. Thus, temperature, pH, degree of agitation, oxygen concentration in the medium and other factors may have to be kept constant during the process. The provision of such conditions requires careful monitoring (data acquisition and analysis) of the fermentation so that any deviation from the specified optimum might be corrected by a control system. Criteria which are monitored frequently are listed in Table 8.1, along with the control processes with which they are associated. As well as aiding the maintenance of constant conditions, the monitoring of a process may provide information on the progress of the fermentation. Such information may indicate the optimum time to harvest or that the fermentation is progressing abnormally which may be indicative of contamination or strain degeneration. Thus, monitoring equipment produces information indicating fermentation progress as well as being linked to a suitable control system.

In initial studies the number of functions which are to be controlled may be restricted in order to gain more knowledge about a particular fermentation. Thus, the pH may be measured and recorded but not maintained at a specified pH or the dissolved oxygen concentration may be determined but no attempt will be made to prevent oxygen depletion.

Also, it is important to consider the need for a sensor and its associated control system to interface

with a computer (to be discussed in a later section). This chapter will consider the general types of control systems which are available, specific monitoring and control systems and the role of computers. More information on instrumentation and control has been written by Flynn (1983, 1984), Armiger (1985), Bull (1985), Rolf and Lim (1985), Bailey and Ollis (1986), Kristiansen (1987), Montague *et al.* (1988), Dusseljee and Feijen (1990), Atkinson and Mavituna (1991) and Royce (1993).

It is apparent from Table 8.1 that a considerable number of process variables may need to be monitored during a fermentation. Methods for measuring these variables, the sensors or other equipment available and possible control procedures are outlined below.

There are three main classes of sensor:

1. Sensors which penetrate into the interior of the fermenter, e.g. pH electrodes, dissolved-oxygen electrodes.
2. Sensors which operate on samples which are continuously withdrawn from the fermenter, e.g. exhaust-gas analysers.
3. Sensors which do not come into contact with the fermentation broth or gases, e.g. tachometers, load cells.

It is also possible to characterize a sensor in relation to its application for process control:

1. *In-line sensor.* The sensor is an integrated part of the fermentation equipment and the measured value obtained from it is used directly for process control.
2. *On-line sensor.* Although the sensor is an integral part of the fermentation equipment, the measured value cannot be used directly for control.

TABLE 8.1. *Process sensors and their possible control functions*

Category	Sensor	Possible control function
Physical	Temperature	Heat/cool
	Pressure	
	Agitator shaft power	
	rpm	
	Foam	Foam control
	Weight	Change flow rate
Chemical	Flow rate	Change flow rate
	pH	Acid or alkali addition, carbon source feed rate
	Redox	Additives to change redox potential
	Oxygen	Change feed rate
	Exit-gas analysis	Change feed rate
	Medium analysis	Change in medium composition

An operator must enter measured values into the control system if the data is to be used in process control.

3. *Off-line sensor.* The sensor is not part of the fermentation equipment. The measured value cannot be used directly for process control. An operator is needed for the actual measurement (e.g. medium analysis or dry weight sample) and for entering the measured values into the control system for process control.

When evaluating sensors to use in measurement and control it is important to consider response time, gain, sensitivity, accuracy, ease and speed of calibration, stability, reliability, output signal (continuous or discontinuous), materials of construction, robustness, sterilization, maintenance, availability to purchase and cost (Flynn, 1983, 1984; Royce, 1993).

METHODS OF MEASURING PROCESS VARIABLES

Temperature

The temperature in a vessel or pipe is one of the most important parameters to monitor and control in any process. It may be measured by mercury-in-glass thermometers, bimetallic thermometers, pressure bulb thermometers, thermocouples, metal-resistance thermometers or thermistors. Metal-resistance thermometers and thermistors are used in most fermentation applications. Accurate mercury-in-glass thermometers

are used to check and calibrate the other forms of temperature sensors, while cheaper thermometers are still used with laboratory fermenters.

MERCURY-IN-GLASS THERMOMETERS

A mercury-in-glass thermometer may be used directly in small bench fermenters, but its fragility restricts its use. In larger fermenters it would be necessary to insert it into a thermometer pocket in the vessel, which introduces a time lag in registering the vessel temperature. This type of thermometer can be used solely for indication, not for automatic control or recording.

ELECTRICAL RESISTANCE THERMOMETERS

It is well known that the electrical resistance of metals changes with temperature variation. This property has been utilized in the design of resistance thermometers. The bulb of the instrument contains the resistance element, a mica framework (for very accurate measurement) or a ceramic framework (robust, but for less accurate measurement) around which the sensing element is wound. A platinum wire of 100- Ω resistance is normally used. Leads emerging from the bulb are connected to the measuring element. The reading is normally obtained by the use of a Wheatstone bridge circuit and is a measure of the average temperature of the sensing element. This type of thermometer does have a greater accuracy ($\pm 0.25\%$) than some of the other measuring devices and is more sensitive to small temperature changes. There is a fast response to detectable changes (1 to 10 seconds), and there is no restriction on distance between the very compact sensing point (30×5 mm) and the display point of reproducible readings. These thermometers are normally enclosed in stainless-steel sheaths if they are to be used in large vessels and ancillary equipment.

THERMISTORS

Thermistors are semiconductors made from specific mixtures of pure oxides of iron, nickel and other metals. Their main characteristic is a large change in resistance with a small temperature change. The change in resistance is a function of absolute temperature. The temperature reading is obtained with a Wheatstone bridge or a simpler or more complex circuit depending on the application. Thermistors are relatively cheap and have proved to be very stable, give reproducible readings, and can be sited remotely from the read-out

point. Their main disadvantage is the marked non-linear temperature versus resistance curve.

TEMPERATURE CONTROL

The use of water jackets or pipe coils within a fermenter as a means of temperature control has been described in Chapter 7. In many small systems there is a heating element, 300 to 400 W capacity being adequate for a 10-dm³ fermenter, and a cooling water supply; these are on or off depending on the need for heating or cooling. The heating element should be as small as possible to reduce the size of the 'heat sink' and resulting overshoot when heating is no longer required. In some cases it may be better to run the cooling water continuously at a steady rate and to have the heating element only connected to the control unit. This can be an expensive mode of operation if the water flows directly to waste. For small-scale use, Harvard Apparatus Ltd (Fircroft Way, Edenbridge, Kent, U.K.) make a unit, the Thermocirculator, which will pump recirculating thermostatically heated water through fermenters of up to 10 dm³ capacity and give temperature control of $\pm 0.1^\circ$.

In large fermenters, where heating during the fermentation is not normally required, a regulatory valve at the cooling-water inlet may be sufficient to control the temperature. There may be provision for circulation of refrigerated brine if excessive cooling is required. Steam inlets to the coil and jacket must be present if a fermenter is being used for batch sterilization of media.

Low agitation speeds are often essential in animal cell culture vessels to minimize shear damage. In these vessels, heating fingers can create local 'hot-spots' which may cause damage to cells very close to them. Heating jackets which have a lower heat output proportional to the surface area (water or silicone rubber covered electrical heating elements — Chapter 7) are used to overcome this problem.

Flow measurement and control

Flow measurement and control of both gases and liquids is important in process management.

GASES

One of the simplest methods for measuring gas flow to a fermenter is by means of a variable area meter. The most commonly used example is a rotameter, which consists of a vertically mounted glass tube with

an increasing bore and enclosing a free-moving float which may be a ball or a hollow thimble. The position of the float in the graduated glass tube is indicative of flow rate. Different sizes can cater for a wide range of flow rates. The accuracy depends on having the gas at a constant pressure, but errors of up to $\pm 10\%$ of full-scale deflection are quoted (Howe *et al.*, 1969). The errors are greatest at low flow rates. Ideally, rotameters should not be sterilized and are therefore normally placed between a gas inlet and a sterile filter. There is no provision for on-line data logging with the simple rotameters. Metal tubes can be used in situations where glass is not satisfactory. In these cases the float position is determined by magnetic or electrical techniques, but this provision has not been normally utilized for fermentation work. Rotameters can also be used to measure liquid flow rates, provided that abrasive particles or fibrous matter are not present.

The use of oxygen and carbon dioxide gas analysers for effluent gas analysis requires the provision of very accurate gas-flow measurement if the analysers are to be used effectively. For this reason thermal mass flowmeters have been utilized for the range 0 to 500 dm³ min⁻¹. These instruments have a $\pm 1\%$ full-scale accuracy and work on the principle of measuring a temperature difference across a heating device placed in the path of the gas flow (Fig. 8.1). Temperature probes such as thermistors are placed upstream and downstream of the heat source, which may be inside or outside the piping.

The mass flow rate of the gas, Q , can be calculated from the specific heat equation:

$$H = QC_p(T_2 - T_1)$$

where H = heat transferred,
 Q = mass flow rate of the gas,

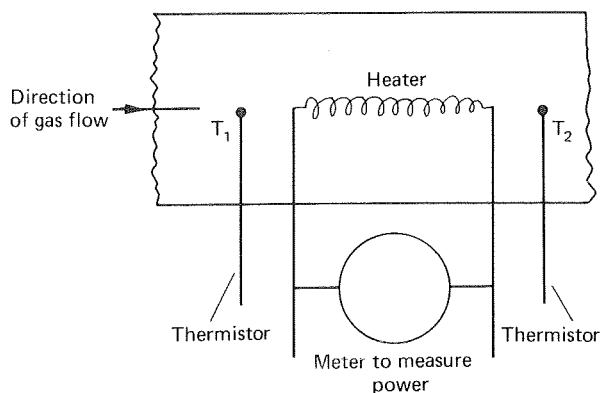


FIG. 8.1. Thermal mass flowmeter.